

Review article

A biophysical approach to cancer dynamics: Quantum chaos and energy turbulence



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ABSTRACT

Cancer is a term used to define a collective set of rapidly evolving cells with immortalized replication, altered epimetabolomes and patterns of longevity. Identifying a common signaling cascade to target all cancers has been a major obstacle in medicine. A quantum dynamic framework has been established to explain mutation theory, biological energy landscapes, cell communication patterns and the cancer interactome under the influence of quantum chaos. Quantum tunneling in mutagenesis, vacuum energy field dynamics, and cytoskeletal networks in tumor morphogenesis have revealed the applicability for description of cancer dynamics, which is discussed with a brief account of endogenous hallucinogens, bioelectromagnetism and water fluctuations. A holistic model of mathematical oncology has been provided to identify key signaling pathways required for the phenotypic reprogramming of cancer through an epigenetic landscape. The paper will also serve as a mathematical guide to understand the cancer interactome by interlinking theoretical and experimental oncology. A multi-dimensional model of quantum evolution by adaptive selection has been established for cancer biology.

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1. Introduction

Cancer is defined as the malignant transformation of tissues characterized by abnormal cell division through the accumulation of unrepaired driver mutations and self-sufficient growth signals. It is a highly complex biosystem that rapidly adapts to its environment, where a single tumor can have hundreds of mutated genes. Due to tumor heterogeneity, personalized pharmacogenomics must be approached as each cancer has a unique signaling identity or a reprogramming cascade for all cancers must be identified.

On a systemic level, cancer is a disease with patient-specific cell-surface biomarkers and gene expression profiles. However, at

a cellular basis, cancer cells outperform normal cells in terms of survival pathways and longevity. Hereby, I will establish the mathematical foundations for the understanding of cancer evolution as a coherent feedback system between the genome, epigenome and the cell-cell/cell-matrix interactome, from which a proteomic landscape will be constructed for reprogramming cancer cells to mutually coexist with non-cancerous cells (i.e. phenotype reversal). To understand these processes, we must explore tumorigenesis and cancer progression under a quantum dynamical framework.

Evolution is currently seen as the variation in the genetics of biological populations over time. However, epigenetic evolution and acquisition of epimutations (genetically non-inheritable changes that determines cell phenotype and gene expression patterns) is not accounted for Darwinian models. The mechanisms by which

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transgenerational epigenetic memory can be transmitted to successive generations, especially over a large period of time must be investigated to understand cancer evolution. More importantly, we must validate whether these on/off gene expression states can be described as quantum states (i.e. qubits). With currently existing experimental evidence and theoretical models, I propose the following hypothesis: *The evolution of cancer cells through quantum selective adaptations to the environment.*

The hallmarks of cancers such as unlimited replication, reprogrammed respirasome, apoptosis evasion, angiogenesis, insensitivity to anti-growth signals, contact inhibition, rewired metabolic cascades, immune system hijacking, etc. have been established by Hanahan and Weinberg (2011). Cancers can selectively over-express or down-regulate protein complexes (ex: cell surface adhesion molecules, membrane proteins, transport molecules, epigenetic factors, enzymes, ion channels, receptors and growth factors), secretory vesicles, exosomes, ribosomes, and lipids depending on their survival needs, adapted mutations and epigenetic changes (Weinberg, 2007). Even amidst aerobic conditions (ample oxygen supply), most cancer cells express hypoxia inducing signaling pathways and the Warburg effect (glycolysis) as an evolutionary strategy to enhance their survival/longevity and proliferation. The excess lactate production allows faster incorporation of carbon into cell biomass and fuels rapid cell division. The suppression of oxidative phosphorylation may help cancer cells evade apoptotic pathways and ageing (energy depletion).

One critical step in oncogenesis involves the up-regulation or reactivation of telomerase in order to overcome the Hayflick limit. Approximately 85–90% of all tumor biopsies are telomerase positive. Although the remainder lack detectable enzyme activity, they can maintain telomere length by ALT (alternate lengthening of telomeres). Furthermore, Cancer stem cells and embryonic stem cells have a depolarized plasma membrane potential in order to proliferate easily and alter their morphogenesis (cytoskeleton) constantly (i.e. regenerative stem cell potency).

Cancers evolve by tissue microenvironment co-dependent phenotype selection. In principle, since cancer cells have a stronger resonant cell-matrix coupling strength than healthy cells (environmental dominance), it is inferred that they can selectively adapt the epigenetic landscape of their microenvironment in favor of their survival cues. According to the proposed model, the temporal evolution of cancer cells' quantum-adaptive energy (epigenetic) landscapes determine morphogenetic plasticity and differentiation through the signaling network dynamics of a perturbation (mutation) induced shift in the underlying pilot wave potential Q . As a result of the altered phenotypic landscape, selective evolutionary properties such as hyper-proliferation and metastatic invasion are acquired by cancer cells via coupling with the energy fields of the tumor microenvironment.

The coupling strength of cell–cell and cell-matrix interactions can be measured using techniques such as force spectroscopy and Bio-AFM (atomic force microscopy) although it varies from cell to cell depending on the environment and measurement context (ex: focal adhesions, cytoskeletal proteins, integrin mediated forces, cell type, etc.). At thermal scales, fluid dynamical models and Boltzmann distributions elucidate cancer dynamics. However, in contrast to the widely held scientific paradigm, the energy dynamics of cancer cells (signaling pathways and information flow) display equivalent topological flux patterns at all orders of magnitude whether at quantum or physiological scales. Mathematically, the energy turbulences of epigenetic landscapes modeling cancer evolution are indistinguishable from the quantum fluctuations governing cosmic inflation.

Empirical evidence asserts that the metastatic potential and cancer cell properties (proliferation, longevity, etc.) are a result of the altered fluid dynamics of cancer's developmental (epigenetic)

landscape. Cancer cells quickly adapt to environmental changes and reorganize their morphology by continuously updating their cell–cell and cell-matrix coupling strengths (chaotic adhesion profile). In theory, these matrix-coupling dynamics can be studied as physiological quantum phenomena by modeling the cancer system as a Bose-Einstein condensate and by measuring the changes in the cytoskeletal-bioelectric field potential or transcriptome-epigenome flux rates.

Epigenetic memory transfer determines cell phenotype and stem cell-lineage specification. The epigenetic modifications determining cancer stem cell reprogramming and the tumor microenvironment's epigenetic reprogramming circuitry have been discussed by many. Epigenetic reprogramming can reset the rate of biological ageing clocks (Rando and Chang, 2012). Altering the DNA methylation patterns alone can lead to stable phenotypic reprogramming of cancer cells (Blancafort et al., 2013). Cell phenotype is the product of an inherited genotype and the many microenvironmental influences (epigenome). Natural Selection (NS) non-randomly selects phenotypes within populations (Reece et al., 2010). The morphological plasticity of cells is governed by the epigenome whereby selective mutations can enhance reproductive success and survival as seen in cancer cells. As cancer cells rapidly divide (altered chronobiology), new mutations that confer selective advantages for microenvironmental adaptation and reproductive success are acquired in subsequent populations.

Apart from covalent post-translational modifications (ex: acetylation, methylation, phosphorylation), extracellular matrix components, cytoskeletal remodeling factors, cell surface adhesion molecules, internal membrane proteins, enzymes, etc. dynamically repress or activate chromatin sites for transcription. For instance, the nuclear lamina interaction with chromatin is essential for normal tissue-specific transcription of certain genes. Hence, nuclear lamina should comprise of epigenetic regulation. By definition, all genetically non-inheritable changes contributing to the cell phenotype that modifies its gene expression and regulation is an epigenetic process/modification (Fig. 1). For instance, glycosylation is proposed as a quantum biological mechanism (Lauc et al., 2014). In argument, epigenetic inheritance can be interpreted as biochemical Lamarckism (Wintrebert, 1962). However, the principal query is whether these epigenetic alterations are macroscopic (biomolecular) quantum systems (i.e. superposition of histone modifications, wave-particle behavior of nucleosome interactions, quantum tunneling of epigenetic signals, etc.)?

Quantum wave effects occur in proteomics. Quantum vibrational free energies of many enzyme complexes such as isomerase and dehydrogenase reactions have been validated. Luo and Lu (2011) verified that the quantum transition between conformational torsion states of polypeptides determines its temperature-dependent protein folding. Quantum coherence in physiological time scales has been well documented. Quantum effects such as the superposition principle and wave-particle duality in photosynthesis cascades (excitons of the chromophores and energy transport complexes), the magnetoreception of birds (quantum entanglement of retinal photoreceptors with the electron spin of Earth's magnetic field) and the olfactory receptors have been experimentally established (Arndt et al., 2009; Lambert et al., 2012; Mohseni et al., 2014).

Schoenlein et al. (1991) validated that vision is a quantum process whereby photon signals are converted to action potentials. The femtosecond isomerisation of rhodopsin via photon interaction is clear evidence to biological quantum information processing. Additionally, quantum wave effects occur in the electron/proton transfers of protein complexes. Electron transfer is gated by environmental energy fluctuations. In theory, oxidative phosphorylation, proton gradients and mitochondrial-metabolome energy transduction should exhibit the quantum phenomena observed in

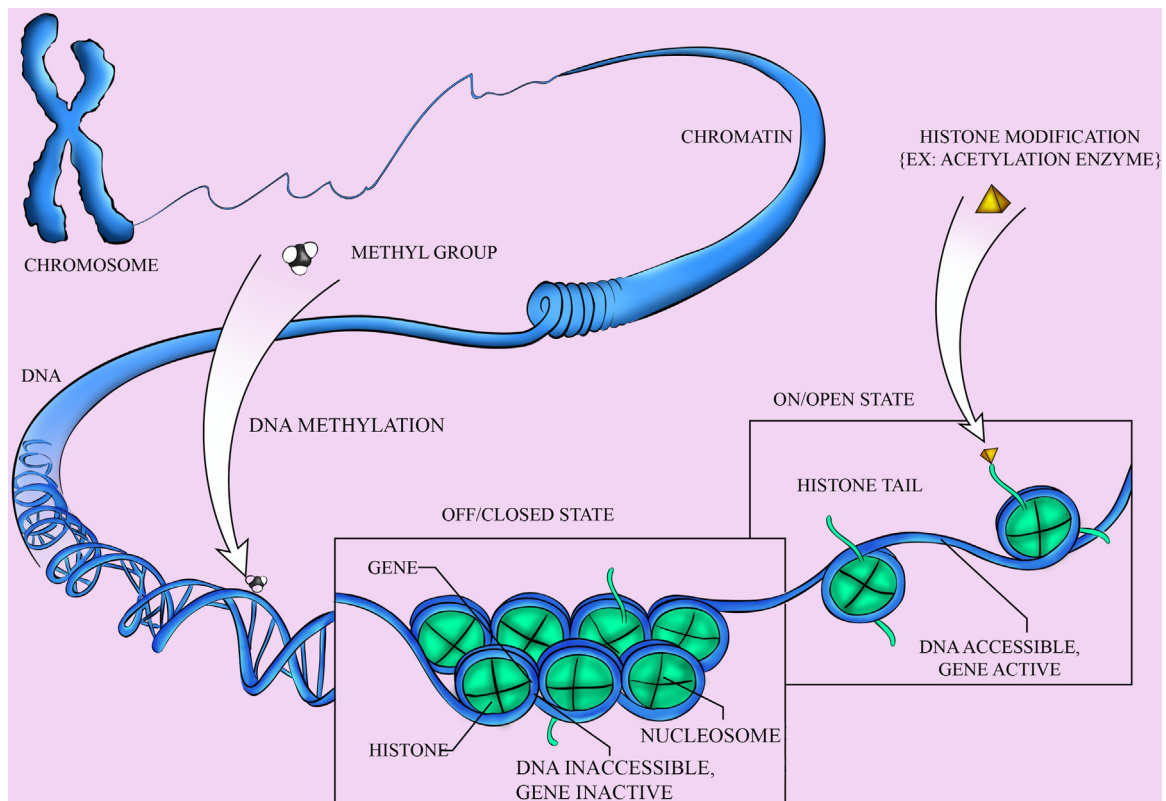


Fig. 1. The Epigenome: A simplified model of the epigenome showing how the DNA's genes are silenced or activated by protein complexes and epigenetic factors resulting in the observed cell phenotype.

photosynthetic systems. The quantum coherence between excitons and charge-transfer states (vibrational modes) of the photosynthetic system can be measured using 2D Fourier transform electronic spectroscopy. (Engel et al., 2007; Romero et al., 2014) The vibrations in the antenna system of photosynthetic bacteria and its light harvesting complexes (ex: FMO protein) are quantum dynamical processes where wave-like energy transfers extends over many pigment molecules through quantum coherence (Sumi, 2001; Fleming et al., 2011).

Turin (1996) derived the vibration theory of olfaction. In contrast to the classical lock and key model of enzyme kinetics, the theory postulates that the odorant molecule once bound to the olfactory receptor's binding site can quantum tunnel through receptor triggering signal transduction pathways. The molecular vibrations of the signaling molecule depending on its electron resonance energy exhibits quantum wave cascades through inelastic electron tunneling. Turin evinced that such quantum leaps occur in the signal transduction of GPCRs (G-Protein Coupled Receptors) depending on their vibrational frequencies and the quantum coherence of signaling molecules. Thus, two molecules with different conformations and chemical structures can elicit identical smells if their bonds vibrate with similar energies.

Such quantum effects describe the radical pair mechanism of avian magnetoreception, whereby the quantum spin of the surrounding magnetic field couples with the retinal cryptochromes (Rodgers and Hore, 2008). Hore further suggests weak magnetic fields may affect the quantum coherence of biochemical reactions (ex: ATP-enzyme kinetics) (Hore, 2012). Even amidst noise, the internal and external magnetic-cryptochrome couplings allow a quantum coherence decay rate in the order of 10^4 s^{-1} (Gauger et al., 2011). Ritz et al. (2004) affirmed that avian magnetoreception consists of quantum effects manifested at the nanoscale. The quantum

Zeno effect is proposed as an explanation to the quantum coherence in magnetoreception (Kominis, 2009).

In principle, not only should such quantum wave effects extend to the mammalian sensory-neural systems but it should apply to other nanoscale biological information processes (e.g. chromatin regulation, histone dynamics, enzyme kinetics, signal transduction pathways, etc.). Spatiotemporal pattern formation in perceptual information processing, morphogenesis and evolution of biosystems can be analyzed within a quantum framework (Igamberdiev and Shklovskiy-Kordi, 2017). Hackermuller et al. (2004) and Gerlich et al. (2011) demonstrated using Talbot-Lau interferometry that large biomolecules produce quantum effects and wave-interference patterns (de-Broglie waves). Small Bose-Einstein condensates were observed in organic macromolecules. Ouyang and Awschalom (2003) studied quantum spin transfer between quantum dots connected by benzene rings (mimics the electron resonance in aromatic amino acids) and observed superposition of spin states. Quantum mechanical measurements explain enzyme kinetics, protein dynamics and biological evolution (Pattee, 1967).

Therefore, I propose that quantum effects should be considered at the level of signal transduction dynamics, protein folding, chromatin regulation, cytoplasmic transport of metabolites, cytoskeletal remodeling, bioelectric- ion channel networks, embryonic development, cell differentiation/morphogenesis, epigenetic modifications, etc. to name a few. The chaotic cancer cell adhesion dynamics and fluctuant membrane fluidity result in rapidly adaptive, fluid-like cytoskeletal rearrangements leading to cancer invasion. Thereby, the morphogenetic plasticity of tissues can be modeled as a non-equilibrium epigenetic landscape undergoing turbulent flux. As such, the cell-matrix interactome is an epigenetic interface to validate the QEH (Quantum Evolution Hypothesis) initially proposed by Ogryzko (1997) and McFadden

(2000). To reprogram cancer cells or any tissue's pathological state to its pre-disease state, a quantum signaling interactome must be conceived. A brief mathematical description of the *cancer reprogramming energy landscape* will be discussed in this paper.

Moreover, 80% of pediatric cancers have metastasized in children at initial diagnosis. Therefore, the immediate diagnostic procedure should incorporate sequencing the entire normal genome and cancer DNA to detect the transformative changes. Direct sequencing alone may not be sufficient in assessing hemizygous chromosomal insertions/deletions (LOH events) and rearrangements at the chromosomal level. Although cytogenetic techniques are widely used in current clinical practice, a *complete genome sequencing and epigenetic profiling* must become diagnostic gold standards.

2. Mutation theory

Cancer is the co-evolution (bidirectional communication) of neoplastic cells and their microenvironment. It is a complex open dynamical system entangled with its environment. Thus, under a quantum perspective, the measurement and the observer must be treated as a single system (i.e. system observing itself). Hence, an 'epigenetic code' accounting for transgenerational epigenetic memory transmission between cells would define the 'Observer' that collapses the biological interactome towards the measured state.

The Human DNA has $>3 \times 10^9$ base pairs. Considering 4 bases, $4^{3 \times 10^9}$ possible position states exist, which is greater than the # of particles in the observable universe. The non-coding regions constituting near 98% of the genome regulates gene expressions, consists of introns, telomeres and derives non-coding RNA. Near half the mitotic chromosome's composition is a non-chromatin sheath of periphery (a potential target for screening epimutations). Meanwhile, the epigenome modulates the changes in chromatin structure in response to noise and environmental signals. Thereby, it acts as a bidirectional switch that can turn on/off genes. It regulates the temporal evolution of gene expression profiles, determines cell phenotype and is the interface between the DNA and environment. ~70% CpG pairs are methylated in somatic cells via DNMTs in the proximal and regulatory regions of the DNA. In cancer cells, genomic hypermethylation often occurs in CpG islands in gene regions while hypomethylation occurs in highly and moderately repeated DNA sequences. Moreover, abnormalities in retrotransposons and chromosomal translocations are seen in cancers. For instance, the inhibition of a single DNA damage response component (DNAPK kinase) can increase DNA translocation nearly 10-fold.

A mutation is a perturbation in the initial set of a cell's information content that iterates within every replication cycle. Darwin (1859) claimed natural selection adaptively selects the fittest phenotype with favorable traits that confer higher reproductive success and survival in a given environment (i.e. Descent with Modification). Many new gene variants can be produced in short time periods in rapidly reproducing organisms. Meanwhile, transgenerational epigenetic inheritance seems to agree with the discredited Lamarckism. The Luria–Delbrück experiment shows that mutations occur in the absence of selection. While mutagenesis seems random (unpredictable), carcinogenesis is the selective (non-stochastic) adaptation to the environment. Hereby, I propose that cancer cells evolve to their environmental fluxes through *quantum adaptive selection* (QEH).

DNA damage is very frequent and at sites of DNA repair, epigenetic changes can remain (Lahtz and Pfeifer, 2011). Most cancer cells have lost one or more genome maintenance and repair systems through mutations, due to which many additional mutations

accumulate. Many years may be required for the acquisition of multiple mutation networks to form a tumor. DNA proofreading, nucleotide excision and mismatch repair mechanisms fix 99% of all DNA errors yet the rate of spontaneous mutations increase as cells undergo ageing and inflammation. Thus, misfolded proteins and altered metabolic signaling are aberrant features of cancer and ageing. Even in the absence of genetic mutations, (epigenetic) changes that alter the regulatory effects or pre-existent patterns of signaling molecules can be described as 'oncogenic' implying cancer is a behavioral complication.

Mutations can occur due to enzymes (ex: nucleases) in the replication process or due to external factors (exposure to mutagens, chemical carcinogens, etc.). Scattering theory describes radiation-dependent mutagenesis. The interaction of a photon (ex: UV irradiation) colliding with DNA can cause inelastic scattering (Raman effect) while the statistically frequent behavior is elastic scattering. Resonance Raman Spectroscopy can be used for the detection of point mutations in cancers by treating them as local plasmonic disorders. Then, the energy derivative of the scattering phase shift can calculate the onset of mutagenesis.

At low energies, the photoelectric effect is given by: $E_{\max} = h\nu - \phi$, where $E_{\max}$ is the energy of the ejected electron, ν is the frequency of the incident photon and ϕ is the work function (minimum energy to delocalize an electron from the surface). At higher energies the Compton Effect dominates and explains the irradiation of DNA (ionizing radiation). The wavelength (λ) shift in ionizing radiation by a single electron is given by: $\Delta\lambda = \frac{h}{m_e c} (1 - \cos\theta)$ where θ = scattering angle and m_e = electron mass. Radiation effects may be direct (ex: mutating oncogenes/TSGs) or indirect action (ex: proton/ion irradiation; free radicals cause oxidative stress damage to the DNA).

The genetic code is essentially contained in the arrangements of H-bonds, where GC and AT base pairs differ by one Hydrogen bond. Although many types of mutagenesis exist (ex: frameshift, point, deletion, etc.), Löwdin (1963) demonstrated that the quantum tunneling of a single H-atom can determine gene mutation and explain the spontaneity of cancer mutations. Post-DNA replication, these protons are in non-stationary states implying a probability for quantum jumps. The quantum mutations give rise to tautomeric bases forming with a probability range of 10^{-3} – 10^{-5} . Kimsey et al. (2015) experimentally verified the occurrence of these tautomeric G-T mispairs ~1/10,000 bases copied using NMR techniques. For instance, radiation can excite the proton to upper energy levels whereby it tunnels easily through the potential barrier. Tunneling is described by a time-dependent Schrödinger equation with a double-well potential for the H-bond. Tunneling frequencies in DNA are in the order of 10^{11} /sec (Löwdin, 1963).

The proton wave-function in the double-well potential can be expressed as: $\psi(t) = e^{-iHt/2\pi\hbar}\psi_0$. Using the WKB approximation, the probability of a proton tunneling through a classically forbidden barrier is given as: $P = e^{-2S}$, where the action $S = \frac{2\pi\hbar}{h} \int \sqrt{2m[V(x) - E]}dx$ and V is the classical potential. The tunneling rate of a dissipative system R is given by: $R = A_0 \exp(-S_{\text{eff}}/\hbar)$, where A_0 is a pre-exponential factor or simply as $R = Pv$, where v is the proton frequency.

Parker and Every (1971) calculated that in the A-T ground state, the lowest tunneling level has a proton life time of 0.165×10^{-2} s and an effective tunneling time of $T_{1,2} = 0.346 \times 10^{-4}$ s. The highest A-T state has a life-time of $\sim 0.3656 \times 10^{-2}$ s and a tunneling time of 0.247×10^{-11} s. The critical temperature for proton tunneling through a barrier (ex: GC base pair) is: $T_c = \frac{\sqrt{2\hbar}}{\pi K_B d} \sqrt{\frac{Q}{m}}$ where Q = electric charge. The energy needed to break an H-bond is comparable to the thermal energy order at room temperature (~ 300 K). Due to thermal fluctuations, a base pair may transit between a paired and open state until the environmental fluctuations exceed

a critical vibration frequency for mutagenesis. Altan-Bonet et al. (2003) reported that the closing-opening transition of a single nucleotide base pair is in the time scale of 10^{-6} seconds. Above a critical temperature, the H-bond system is a weak glassy phase. When the environmental temperature T is higher than the glass transition temperature T_g , the folding-unfolding of biopolymers and polypeptides is a two state process. When $T < T_g$, it has many intermediate transitions (spin glass).

Ogryzko (1997) was the first to pioneer the Quantum Evolution Hypothesis (QEH). Given that cell growth in particular environmental conditions is a quantum operator (observable), Ogryzko suggested that cells may behave as a quantum wave function (be in a state of superposition of eigenfunctions of the operator) and upon measurement directed mutants are selected. Hence, varying growth conditions correspond to different superposition and thereby the quantum selection of different mutants (Ogryzko, 1997; Bordonaro and Ogryzko, 2013). McFadden (2000) claimed quantum effects can selectively guide mutagenesis towards adaptive genetic variation. Likewise, the quantum evolution and epi evolution model formulated in this paper states that although mutagenesis appears to be stochastic due to the lack of information on the initial conditions and the resultant chaos in the co-evolution of a biological system and its environment, the selection of these mutations by the biosystem (organism/cells and environment) is a deterministic quantum flux process.

Quantum mechanics endows living organisms with the ability to initiate specific actions, including new mutations (i.e. *Quantum adaptive evolution by selective mutagenesis and deterministic NS*) (McFadden, 2000). Cells may be able to choose to mutate particular genes that provide an advantage in the environment in which the cell finds itself at quantum level wave-field fluctuations (i.e. non-randomly selected adaptive mutations). Under this perspective, an epimutation is a reversibly heritable change in gene activity caused by environmental quantum jitters/fluxes. Given that these epimutations are selective, then morphogenesis and gene expression-regulation dynamics (phenotype plasticity) are quantum deterministic systems.

Djordjevic (2016) demonstrated that quantum mechanical models accurately illustrate the formation of spontaneous, induced and adaptive mutations. Cairns and Foster (1991) discovered that *E. coli* produced adaptive mutations with respect to lactose. McFadden and Al-Khalili (1999) provided a possible mechanism for how *E. coli* developed adaptive mutations to digest lactose at a faster rate than the expected rate for random mutations using quantum theory. Hence, by claiming that quantum effects account for the adaptive mutational variations co-determined by the environment, they further supported Ogryzko's hypothesis.

While Ogryzko's model is on quantum effects at the cellular level, Al-Khalili and McFadden provided a quantum model of adaptation focused at the level of proton tunneling in DNA supporting the works of Löwdin where the superposition of H-bonds sped up the environmental adaptation process. Adaptive mutations may be generated by the environment induced wave-function collapse of the DNA's superposition of mutational states. Two Gaussian wave packets separated by a distance Δx have the decoherence time $t_D = t_R (\lambda_T / \Delta x)^2$, where $\lambda_T = \hbar / (2 m k_B T)^{1/2}$ is the thermal de Broglie wavelength, $\Delta x \approx 0.27 \text{ \AA}$ for a proton at 300 K and t_R is the relaxation time (the speed of energy dissipation due to interaction with environment) (McFadden and Al-Khalili, 1999). Thus, chaotic decoherence of H-bond entanglements caused by the presence of a single environmental resource gradient (lactose in the case above) can accelerate the generation of mutant states.

Davies (2004) provided an in depth review on quantum information processing in biological systems from mutagenesis to the origin of life with experimental suggestions. Furthermore, Davies et al. (2011) described cancer cells using phase transition portraits

and potential energy landscapes satisfying the Klein-Gordon equation: $\frac{d^2 x}{dt^2} = -\frac{dF}{dx}$, where F is the free-energy. Considering 90% of our cells consist of the microbiota, the microbiome's influence on cancer has been well-established. The diet-gut microbiome interactions alter the epimetabolome. The gut microbiota was shown to regulate epigenetic reprogramming in host tissues by altering histone acetylation and methylation levels via short-chain fatty acids (SCFAs) and HDACs. Certain bacterial infections can evade the immune system and produce mutagenic effects on the cytokines released by inflammatory cells further linking its association to carcinogenesis. Although viruses' link with oncogenesis is well known, it must be noted that 30% of all protein adaptations since our divergence from chimpanzees were driven by viruses as well.

All cells read chemical signals from neighboring cells to make collective decisions after exchanging information about their environment. Cells sense concentration gradients of proteins and hormones to have spatiotemporal references for directional growth and motion during morphogenesis. Thus, local immediate surroundings have a causal effect in the malignant transformation of neighboring cells. The collapse of non-local H-bonds' superposition during mutagenesis (as a certain threshold frequency is surpassed) chaotically alters these local cell communication networks.

In a non-isolated system in contact with its surroundings, the quantum coherence will decay with time. Zurek (1991) derived an expression for the time scale over which quantum coherence is lost via environmental coupling. Decoherence explains the loss of information of a system's dynamics through its entanglements with the surroundings (ex: leakage into a heat bath). It eliminates the off-diagonal terms of the observed system's local density matrix and one of the remaining preferred states are selected upon measurement (Schlosshauer, 2004). The multitude of interactions in such heat baths and the macroscopic dimensions (correspondence of many occupied quantum states) of the environment ensure decoherence, as seen by the loss of phase information between state transitions in quantum systems.

Quantum decoherence in a thermodynamic (Boltzmann) system can be approximated as: $t_{\text{coherence}} \sim \frac{1}{mT}$ where m = mass and T = temperature. The amount of entanglement depends on the coupling between biological oscillators which if the interaction energy satisfies the inequality ratio: $\frac{2k_B T}{\hbar \omega} < 1$, quantum effects are present. Based on optical frequencies of the DNA, Rieper et al. (2011) calculated that the ratio is below 1 and thereby, cells are quantum entangled systems at 300 K (room temperature).

However, (de)coherence alone does not explain the measurement problem yet it describes the transition of a system into a mixture of states that the observer selects from. According to QEH, cancer cells select the preferred mutant states (phenotypes) that favor their adaptive evolution (survival and reproduction) indicating a higher resonant-energy coupling to the environment's epigenetic flux landscape in comparison to healthy cells (i.e. stable quantum coherence or $\uparrow$ # of entanglements generated between system and environment).

The coupling interaction between a system and the observer performing the measurement can be defined as the sum of their Hamiltonians (Bohm, 1951). Thence, the perturbed interaction Hamiltonian between cancer cells and the environment must be significantly higher than the Hamiltonian between healthy cells and environment: $\hat{H}_{\text{cancer}} \gg \hat{H}_{\text{healthy}}$. The wave function collapses with the decay of superposition of states into mixtures of preferred states chosen by the environment-induced superselection. Thereby, in extension to the Quantum Evolution Hypothesis, the quantum potential Q guiding cancer cells must have a higher resonance coupling with the environment in order to be the dominant population (Fig. 2).

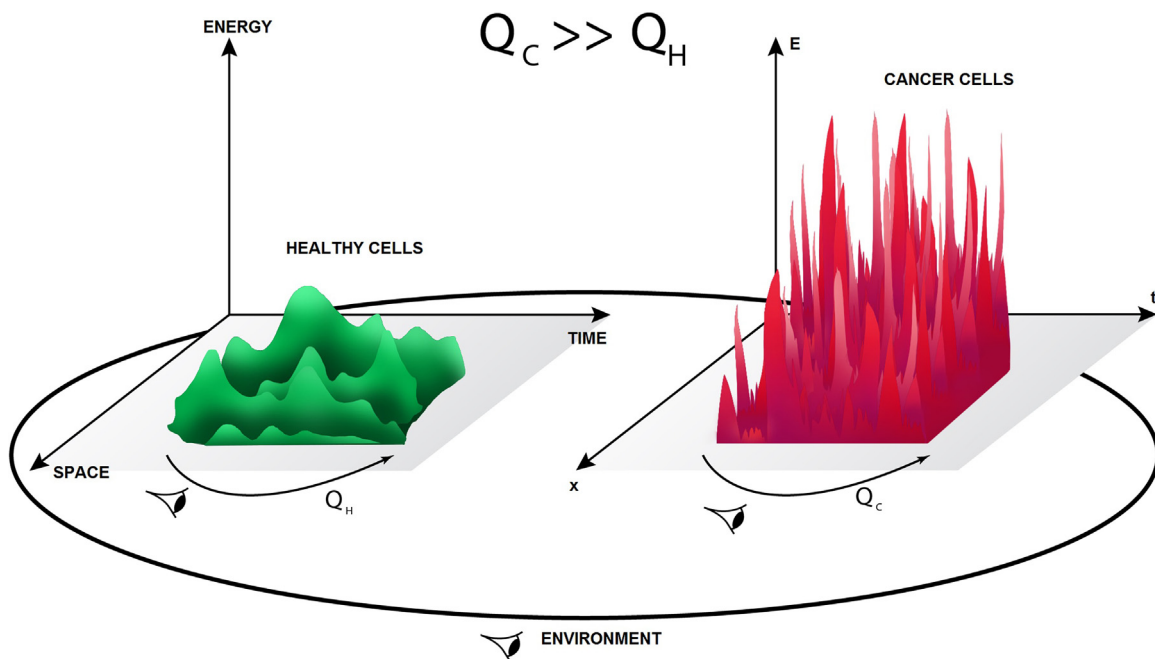


Fig. 2. Quantum Evolution Hypothesis: The epimetabolome of cancer cells has a stronger coupling to the environment than healthy cells as indicated by the strength of the energy fluctuations and the guiding Quantum Potential Q selecting the quantum adaptive state in pre-space.

The thermodynamic irreversibility of a measurement implies destruction of definite phase relations/information deriving Boltzmann statistics and probability distributions to describe the measurement dynamics. Re-establishment of definite phase relations between eigenfunctions of a measured variable is extremely unlikely since the decrease in entropy of a system is highly unlikely (Bohm, 1951). However, biological systems utilize negentropy (increase the entropy of their surroundings to maintain internal order) and cell states are epigenetically reversible.

H-bond networks also form the secondary structures of proteins, misfolded states, dysregulated or selectively altered expressions of which are characteristics of cancer signaling. They are also the basis of water structuring in the cytoplasm and matrix. Hence, coherent proton tunneling between base pairs and disordered H-bond networks (disoriented spin lattices) in cell-protein structures are signatures of tumorigenesis. Asano et al. (2013) further extended adaptive mutations theory towards a quantum epigenome model where epimutations were described in terms of the chromatin's quantum entanglements with the environment.

Vibrationally enhanced enzymatic hydrogen transfer (tunneling) is mediated by thermal fluctuations of the enzymes' active sites (Bruno and Bialek, 1992). For a static barrier, the tunneling probability is expressed as $\rho_{\text{tunneling}} \propto e^{-\frac{2\phi}{\hbar}}$. For a fluctuating barrier, shorter transfer distances occur. Furthermore, a transcriptase can decipher the DNA's H-bond superposition at an interval of $\Delta t \ll 10^{-13}$ s (Cooper, 2009). Cooper further supported the quantum Darwinian evolution model by suggesting that quantum entanglements between coherent protons in amino acids and DNA are qubits. The coupled quantum oscillations of the proton interactions between proteins and DNA forms a quantum molecular clock describing biological information processing at frequencies in the order of $\sim 10^{13}$ Hz.

Moreover, PCR-DNA amplification can probe quantum coherence in the thermodynamic macromolecules of a biological system (Bieberich, 2000). Primer-template duplexes were shown to be in a superposition of binding states since an additional 500 J/mol binding potential was calculated. Hence, Bieberich demonstrated that primer annealing follows Bose-Einstein statistics and fractal

macromolecular structures (proteins and biopolymers) promote quantum coherence of binding sites.

There is an approach called Quantum Bayesianism (QBism) claiming that quantum statistics in biosystems is not necessarily a result of their quantum nature. Rather adaptive dynamics of the wave function collapses due to the observer updating his/her beliefs of the system post-measurement(s) (Asano et al., 2011). Hence, QBism portrays the subjective interpretation of an objective wave function measured by individual observers. Karafyllidis (2008) suggested a quantum model for the information processing between DNA and proteins. Using DNA triplets (codons) as the basic unit of information storage, a 64-dimensional (4^3) Hilbert space was constructed.

As for the measurement problem, many explanations have been proposed with equivalent quantum formalism such as the wave function collapse, Bohr's complementarity principle, Many Worlds interpretation, cyclical cosmology models, de Broglie-Bohm theory, etc. For example, Wheeler's one-electron universe states sub-atomic particles are conscious observers collapsing the wave functions of their environment. In the Many Worlds interpretation, an objective universal wave function exists where nature doesn't decide between outcomes (parallel universes). In Superstring theory, the collisions of vibrating branes create big bang singularities (cyclic universes).

In quantum biological systems, the *quantum Zeno effect* enables coherent superpositions and quantum entanglements to persist in macroscopic time scales (Igamberdiev, 2007). The quantum Zeno effect states that given an unstable (chaotic) quantum system, frequent measurements could inhibit the *exponential decay* (decoherence) of the system. Dyson (2004) claims the mind is inherited in every electron. The observers exist not only within the measured system but externally as well (a Universal observer/wave function) while upholding the Conservation of Energy. The information flux within and out of the observed system (implicate and explicate order) can be described by the flow of energy in a pre-space geometry (Bohm and Hiley, 1993). Then, cancer evolution can be seen as the quantum adaptive selection of a biosystem's dynamical wavefunction in pre-space (Bohm, 1980).

The Universal wave function is a self-organized pilot wave acting as the binary quantum switch, cohering and decohering any measured system and its sub-systems in temporal evolution. The Universal wavefunction with time reversal invariance is given by a Bohmian *guiding equation* describing the evolution of particles' position states (energy-surface): $\frac{dq(t)}{dt} = \frac{\hbar}{m} \text{Im} \left[\frac{\psi^* \partial \psi}{\psi^2} \right] (q_1, \dots, q_N) = \text{Re} \left(\frac{\partial \psi}{\partial t} \right)$, where q is the particle coordinate. The quantum Liouville theorem provides the time evolution of a mixed state density: $\frac{\partial \rho}{\partial t} = \frac{1}{i\hbar} [H, \rho]$.

Therefore, with the epigenome actively observing the environmental perturbations (Maxwell's Demon), living cells are biological quantum computers. The epigenome is the bistable quantum switch regulating the cell's metabolic circuit. The H-atom is the qubit undergoing the Rabi oscillations. Hence, the measurements of the epigenome can be modeled as the inverse quantum Zeno effect of a Rabi cycle's oscillations, forcing the dynamics of the observed system to evolve along its chosen Feynman path. Given a Rabi frequency ω and a qubit (H-bond) that exists in a superposition of states $|0\rangle$ (on) and $|1\rangle$ (off), the time-dependent Schrödinger equation describes the temporal evolution of the Rabi Oscillator: $i\hbar \frac{\partial \psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(x,t) + Q \right] \psi(x,t)$, where Q is the quantum potential. The probability of the qubit being found in state $|1\rangle$ post-resonance is simplified as: $P(t) = \sin^2 \left(\frac{\omega t}{2} \right)$.

Epigenetic mutation rates are at least one or two orders of magnitude greater than somatic DNA mutation rates as the epigenome rapidly responds to the fluctuant environment (Horsthemke, 2006). In principle, since cancer cells generally have higher replication rates than normal cells, epigenetic memory inheritance should accommodate adaptive changes at a faster time-scale and thereby correspond to highly frequent epimutations. Therefore, DNA methylation and histone modification rates are dynamically fluctuant and are difficult to compute. Mapping them will provide a complete description of cancer's evolutionary dynamics (Dobrovic and Kristensen, 2009). In terms of experimental applications, Whole Genome Bisulfite Sequencing (Methyl-C Seq/WGBS) maps the methylome, ChIP Sequencing (Chromatin Immunoprecipitation) maps the histone profile and RNA Sequencing (RNA-Seq) maps the transcriptome profile of cells.

Decoherence in open systems induced by a fluid environment (ex: a Brownian particle) is defined as Einselection (Environment induced super-selection), whereby the energy spacing of the system is large compared to environmental frequencies (Zurek and Paz, 1999). The observation of an actual Brownian particle is the average of the unpredictable forces acting on it. The probability flow of a Brownian particle is given by the Fokker-Planck equations. The time evolution of the particle's probability density function respects the continuity equation: $\frac{\partial P(x,v,t)}{\partial t} = -\nabla \cdot j$, where j is the probability current and v is the velocity. Such probability fluid equations are consistent with Boltzmann equilibrium distributions. However, the averaged solution of the Wiener process neglects the turbulence observed in real biological systems.

The measurement of the dynamic coupling entanglements between an initial state and the environment is given by the von-Neumann entropy: $H(t) = -\text{Tr}(\rho(t) \log \rho(t))$, where ρ is the density matrix. Here, active measurement and collapse is defined in terms of energy. As stated, the measurement dynamics in a quantum biological system can be traced by the epigenome, which unifies the dual states: the observer and that which is observed (environment and DNA/cell), as one active player. In biological systems, inputs from energy sources (environment, mitochondria, ion channels, etc.) are coupled to reactions that aren't entropically favored (Gibbs energy > 0). This coupling allows endergonic reactions like DNA synthesis to proceed by increasing the entropy of the Universe

while maintaining low entropy in the microcosm (living system). Schrödinger was the first to suggest that quantum fluctuations produce mutations in genetic structures.

Schrödinger (1944) coined the term negentropy to describe the self-organization of life through feedback network loops maintained by extracting order from the environment. Negentropy is expressed as $S = -k \ln W$ or $-S = k_B \log(1/D)$, where D is a quantitative measure of atomistic disorder, $W = \#$ of thermodynamic states and k_B = Boltzmann constant. Schrödinger establishes the Mutability formula from Delbruck's Model: $t = \tau e^{E/kT}$, where E is the threshold energy above which quantum molecular vibrations in the DNA result in mutations. τ is a constant in the order of 10^{-13} – 10^{-14} and t = time of expectation for mutation. He further expresses t as a temperature T – dependent ratio: $\frac{t_{T+10}}{t_T} = e^{-10E/kT^2}$. In cancer cells, environmental order (energy) is harnessed to increase topological disorder to enable its metastatic invasion and progression. Therefore, cancer is a rapidly evolving organism employing morphological chaos as a survival strategy. Normal cells have a limit to their internal equilibrium (homeostasis) and internal disorder increases as they age. By violating such spatio-temporal limits, cancer cells have reached permanent negentropy in vitro or until the death of the organism/system in vivo.

The probability of a protein binding to a specific DNA sequence site is governed by its relative binding energy. A single DNA strand resembles a coupled chain of harmonic oscillators, where each 2 neighboring states with distance r have dipole-dipole interaction, a defined Hamiltonian for n bases and a coupling strength ω . If $\omega < 1$, state quantum entanglement exists at room temperature for DNA's information flow. Schrödinger proposed that the environment can be modelled as quantum vortices or a heat bath of harmonic oscillators in every point of space-time interacting with an open (biological) quantum system. The coupling between neighboring electron clouds of nucleic acids is then: $\omega = \sqrt{\frac{Q^2}{4\pi\epsilon_0 m r^3}}$ where $r = 4.5 \text{ \AA}$ and ϵ_0 is the vacuum permittivity (Rieper et al., 2011). Since we are dealing with harmonic genetic oscillators, this coupling constant is simply the angular frequency satisfying the Planck-Einstein relation $E = \hbar\omega$.

3. Tumor morphogenesis

Cancer is the co-evolution of neoplastic cells and its surrounding. Cancer is a disease of the microenvironment; it is microevolution. Fitness of the driver mutations amidst the dynamic microenvironment corresponds to specific epigenetic changes enacting as the tumor's master regulator. The tumor then hijacks the metabolome and energy fields of surrounding cells as a result of which tissue polarity and morphology alters. For instance, the mTOR/elf4E/p53/IGF-1 pathway links cancer metabolome and ageing to environmental resource gradients. These signaling networks are reprogrammed by cancer cells in favor of their survival over the collective organism.

The chaotic cell-matrix couplings and disorganized cytoskeletal remodelling are signatures of tumorigenesis. The cell-matrix adhesion dynamics determine cancer morphogenesis and allow selective metabolic reprogramming. Metastasis accounts for ~90% of cancer-related deaths. Metastatic invasion is determined by microenvironment co-dependent morphogenetic differentiation of cells and the resultant adhesion dynamics. The cytoskeleton refers to the network of actin, microtubules and intermediate filaments, whereas the Extracellular Matrix (ECM) consists of protein complexes, cell adhesion molecules (CAMs), fibroblasts, collagen, fibronectin, laminin, etc. Through ECM signaling, the cytoskeleton provides the structural framework of cells (determines cell shape and cytoplasm's organization) and coordinates the movements of cells, organelles and chromosomes during cell division.

Fröhlich (1968) predicted quantum coherence in living cells due to microtubules (MTs). He postulated that intrinsic biochemical energy and thermal energy from the heat bath of surroundings provides energy for coherent conformational excitations of microtubule protein dipoles in the excitation frequency order of 10^{11} to 10^{12} Hz. Fröhlich defined MTs as pumped phonons and Bose-Einstein condensates (Fröhlich, 1968). If more than a certain threshold of energy is supplied at a constant rate to the coupled molecular oscillators in a heat bath, then all oscillators (cells or protein/gene networks) condense into a giant dipole (a BEC condensate).

The Debye model depicts the vibrations of protein lattices (heat dissipated from spin crystals) as phonons. The net electrostatic force acting on the six surrounding tubulins in a hexagonal lattice

is given as: $F = \frac{e^2}{4\pi\epsilon} \sum_{i=1}^6 y_i/r_i$, where y_i and r_i are inter-tubulin distances and ϵ is the average protein permittivity. Based on a linear chain of tubulins to represent 3D MTs, Reimers et al. (2009) concluded that weak Bose-Einstein condensates vibrating at ~ 8 MHz in the picoseconds range can form in microtubule networks. However, they found no evidence to support Fröhlich condensation in enzyme kinetics.

According to Penrose and Hameroff's Orch OR theory (Orchestrated Objective Reduction) (1996), when space-time in the brain gets sufficiently curved by gravity, a nonlinear instability causes the wave-function collapse at a threshold. Each tubulin can consist of a multiple quantum states due to epigenetic post-translational modifications. Gap junctions mediate quantum tunnelling thresholds whereby cell networks function as Bose-Einstein condensates. Thereby, tubulin qubits form the electrical nervous system of cells, where the information-energy density flux resulting from the quantum vibrations of MT networks defines quantum fractal consciousness. The model coincides with Pribram-Bohm's holographic universe, in which our perceived reality is the result of neural Fourier transforms (*quantum consciousness*).

In the Penrose-Hameroff model, Microtubule Associated Proteins (MAPs) attached to microtubules tune quantum oscillations and results in collapse (Hameroff et al., 2001). Energy is provided by tubulin GTP hydrolysis/binding and MAP phosphorylation. Each tubulin can support quantum level electron delocalization. Samsonovich et al. (1992) showed that stimulation of a 2-D tubulin sheet rolled into torus shows long range Fröhlich resonance matching with experimentally observed MAP attachment patterns.

Orchestrated Reduction is given by the uncertainty relation: $E = \hbar/t$, where t is the coherence time. Libet et al. claimed that the time scale for pre-conscious to conscious transition (self-collapse of tubulin networks' superposition) is ~ 500 ms. Loewenstein (2013) further claims that neural quantum information processing has a decoherence time $\sim 10^{-13}$ – 10^{-20} seconds, whereas physiological time-scales of consciousness are 10^{-2} – 10^{-1} s. Hameroff (2004) further extended the Orch-OR theory to the *origin of cancer*, where the abnormal segregation of chromosomes during mitosis (aneuploidy) is due to the impairment of quantum entanglements among microtubule mitotic spindles, centrioles and the cell cytoskeleton. The hyper-proliferation of cancer cells may be defined as the quantum dynamics of reversible microtubule-actin network remodeling and their corresponding reversible epigenetic modifications (Hameroff, 2004). Thereby, the fluctuant and hyperchaotically adaptive cytoskeletal networks (phenotype/epigenetic cell state) correspond to a rapidly evolving energy landscape (epigenetic inheritance) in cancer cells. The quantum vibrations and the quantum dynamical coherence (non-equilibrium Bose-Einstein Condensate fluctuations) of cytoskeletal-epigenetic dynamics in cancer cells should be measurable by Raman spectroscopy (can be visualized as the fluctuations of protein condensates (spec-

trograms) where discrete Raman peaks merge at certain regions indicating condensation) (Sahu et al., 2013a). In addition to the quantum coherence detected by resonance Raman spectroscopy, I further propose that Bio-AFM and fluorescence spectroscopy can assess the Fröhlich coherence/BEC condensate dynamics of cytoskeletal (microtubules and actin) networks and proteins in cancer cells in room temperature. The quantum entanglements of MT spin-networks are Bose-Einstein Condensates. Hameroff further claims centrioles have vast information storage capacity to determine all possible cell states of differentiation (phenotypes) in an organism. The e^- or p^+ tunneling observed in the condensate, can be modelled by a spin boson system's spectral density function: $J(\omega) = \frac{4\pi}{\hbar} \sum_a C^2(\omega - \omega_a)$ where ω = tunneling frequency.

Hameroff and Penrose (1996) derived the gravitational self-energy for a superpositional tubulin: $E_G = Gm^2/a$, where a is the superposition separation distance (ex: the spherical radius of a carbon atom). Tegmark (2000) derived a decoherence time equation: $t \sim \frac{4\pi\epsilon_0 a^3 \sqrt{mkT}}{Nq_e^2 s}$, where D = tubulin diameter, N is # elementary charges in MT interacting system, g is the Coulomb constant and q = electron charge. Tegmark used 24 nm linear solitons to model the MT-ion collisions to be $\sim 10^{-13}$ s. Meanwhile, the Orch OR theory states that the superposition length 'a' is in the femtometer range (the atomic nuclear radius).

Hagan et al. (2001) confirmed that Tegmark's equation derives a superposition of neural firings and soliton superposition with decoherence times of 10^{-20} s and 10^{-13} s respectively. They further calculated that according to Orch OR's MT superpositions in coherent dipole states and with shielding derives decoherence times of 10^{-4} s and 10^{-2} – 10^{-1} s respectively (i.e. physiologically relevant time scales). Therefore, Hagan et al. assert that neural MTs process qubits and could avoid decoherence via long quantum processing times. They further suggest that Van der Waals forces in biopolymer interactions result from quantum coherence.

Sahu et al. (2013a, 2013b) used 4 nm electrodes to measure resonance fractal repeats in several orders of magnitude in MTs. Topological qubits are quantum fluxes at millikelvin temperature where information is encoded in space (energy-surface) and is selective of phases. Phase change measurements of an AC signal between 2 ends of the MT measured peaks statistically following a very certain trend in multiples of 45° indicating a non-random detection of topological qubits (quantum phase leaps) following Fibonacci geometry. MTs were shown to store and process qubits without releasing or consuming heat (solitary BECs). Furthermore, MT conductivity was determined to be independent of thermal fluctuations (signaling remains unaffected by thermal noise). Matsuno (1999) suggested that ATP hydrolysis and actin-myosin ATPase activities in cell motility are quantum coherent energy processes.

Cytoskeletal polymerization is defined by Michaelis-Menten kinetics: $\frac{dn}{dt} = k_{on} (C_0 - M \frac{n(t)}{V}) - k_{off}$ where V is the volume, M = # nuclei and the right-side term excluding the $-k_{off}$ is the growth of the cytoskeletal filaments. Activator-repressor models of genetic network oscillations can be described classically by equilibrium rate equations (particles) or reaction-diffusion models (waves) given the binding energies (affinities) and concentration of the complexes.

As mentioned, cancer cell adhesion pathways' influence on the metastatic profile has been strongly established. For instance, the Focal Adhesion Kinase (FAK)/Crk/Src pathway regulates cancer cell –matrix coupling dynamics (Uthamacumaran et al., 2009). Cell-matrix adhesion assays can map the phosphorylation patterns of these signaling cascades as an indirect measurement of post-translational modifications (epigenetic control) on the cancer transcriptome. Bio-AFM microscopy and single molecular fluores-

cence techniques can assess the adhesion strength of cells and provide a quantifiable metastatic landscape.

Ramis-Conde et al. (2008) modeled cancer cell invasion as the cell migration dynamics determined by chemotactic gradients and focal adhesion complexes arising from the degraded ECM. Thereby, cells can be visualized as interacting free particles. The motion of cells is described by a mitosis rate given as: $R(x_i, t) = f(A(x, t) + p_0)$ and a mitosis probability given by: $f(A(x, t)) = \left(\frac{\tau A(x, t)}{\tau A(x, t) + 1}\right) P$, where τ is the instantaneous reaction rate between growth factor receptors and $A(x, t)$ is the density of the degraded matrix which cells can absorb. Growth factors (GFs), hormones and signaling metabolites from. Such simple models depict how waves of chemo-attractants diffuse in space and derive heterogeneous concentration profiles regulating the spatiotemporal evolution and morphological differentiation of cells.

At the transcriptome level, the coupling strength of cancer cells can be defined by the frequency of its genetic oscillators ω_i . For instance, the phases of two heterogeneous cell signaling oscillators following Michaelis-Menten type coupling is given by: $\frac{d\theta}{dt} = \omega_i + \mu H(\theta' - \theta)$ and the same equation holds true for ω_i' , given ω and ω' are the 2 coupled oscillators' frequencies, μ = chemical potential, T is the period and the average interaction function $H(\theta' - \theta)$ describes the phase change Hamiltonian due to the coupling signal. $\Delta\omega_i = \theta' - \theta$.

Meanwhile, epigenetic states can be modeled as bistable solutions of a bifurcation diagram (Sedhigi and Sengupta, 2007). Sedhigi claimed that the epigenetic system's state evolution depends on hysteresis (i.e. its history). Under a quantum dynamical context, this implies that epigenetic evolution of cells conform the 'sum over histories approach' (path integral formulation) and sensitive dependence on initial conditions (Bohmian chaos). The bifurcations should not be seen as separate branches but as the superposition or wave interference between two entangled states. The potential evolutionary trajectories of a single attractor are in superposition until selection and the temporal evolution of the turbulent attractors are defined by qubits.

Waddington (1942) defined epigenetics as the branch of biology which studies the causal interactions between genes and their products bringing phenotype into being. He formulated the *epigenetic landscape*, a dynamic formalism that models biological development via self-assembly, co-exchange of energy with surroundings and autonomous regulation of internal order amidst systemic chaos. Epigenetic changes enable the Waddington landscape to unfold cellular differentiation pathways (morphogenesis), where cell fates are attractor states in the lowest points of the valleys (local optima in basins of attraction). Under the QEH and Bohm's Interpretation of QM, the Waddington landscape becomes a *deterministic non-local quantum adaptive development (epigenetic landscape)* (Fig. 3).

Cancer cells are highly fractal topologies. The hydrodynamics of cancer cells motion and energy metabolomes are more turbulent (higher Reynolds number) than normal cells. Thereby, cancer cell epigenetic states are chaotic attractors (i.e. strange attractors/fractals, oscillatory vortices, unstable limit cycles or non-equilibria points) with turbulent energy dynamics around and in between (quantum) state transitions. There's a bidirectional interchange between heredity and the environment through the epigenetic landscape as demonstrated by transcription factors- induced *time-reversibility* of cell fates in Yamanaka's induced Pluripotent Stem Cell reprogramming (iPSCs) (Yamanaka et al., 2007). Phenotype (epigenetic) reprogramming has not been achieved in the context of cancer stem cells although they express the Yamanaka circuits.

During iPSC reprogramming, almost complete demethylation of the Oct-4, Nanog and Sox-2 promoters have been reported implying

epigenetic control of cellular reprogramming. However, it is a late step since cells that acquired self-renewal still shows high levels of DNA methylation. Transcription Factors (TFs) constitute around 6% of the 26,383 genes analyzed in the human genome by Venter et al. (2001). Furthermore, 902 proto-oncogenes were identified in the analyzed genome and 23 of them are conserved in vertebrates and invertebrates. TFs Oct 4, Sox2 and Klf4 access silent chromatin sites and initiate cell fate changes via DBDs (DNA Binding Domains). Such epigenetic processes form a biological clock along with the signaling frequencies of the cell interactome and the circadian rhythm. Graf and Enver (2009) discussed a phenotypic landscape for reprogramming cell lineages through such transcription factors.

Turing (1952) coined the term morphogen, the diffusion of which establishes the *pre-pattern* of biological development. Interestingly, the same year Bohm developed the deterministic interpretation of QT on *pre-spaces*. A reaction-diffusion system will induce non-uniform spatial patterns in an open homogeneous system as locally stable states become globally unstable through Brownian diffusion. The diffusion of morphogens and cancer stem cell gradients will exponentially drift from their initial conditions (chaos theory).

Turing further stated that the inhomogeneous biological system behaves like a series of electrical oscillators whose wave amplitudes are governed by the metabolic circuit and the disturbances affecting the frequencies/energy spectrum of signaling cascades determines their oscillation phase evolution. Thereby, in cancers, the default threshold frequency is perturbed in the genetic oscillators governing epimutational-development networks due to quantum adaptive mutations. Cancer cells are highly adaptive non-equilibrium systems where self-organized symmetry breaking (bifurcations) in tissue pattern formation result from the activator-inhibitor dynamics of chaotic attractors (signaling molecules and gene networks).

Turing's pattern formation equations can be simplified by the following 2-component diffusion system:

$$\begin{cases} \frac{\partial u}{\partial t} = \gamma f(u, v) + D_u \nabla_u^2 \\ \frac{\partial v}{\partial t} = \gamma g(u, v) + D_v \nabla_v^2 \end{cases}, \text{ where } \gamma \text{ is a variable scale parameter, } u \text{ and } v \text{ denote the activator and inhibitor.}$$

According to this model, waves of morphogens diffuse across the cell development space forming constructive and destructive interference patterns resulting in an inhomogeneous spatial patterning due to the (quantum selective) resonance or damping of biomolecular concentrations. While Turing developed a pre-patterning model with dynamical instabilities, many other mathematical models for morphogenesis and developmental biology exist. For example, Wolpert (1969) derived the French Flag Model where protein-embryonic gradient thresholds produce fields under different protein intensities to yield different cell phenotypes. Cooke and Zeeman (1976) developed the Clock and Wave front Model where a clock or gene oscillator governing the periodicity of the morphogen pattern interacts with signaling wave fronts.

Through dynamically altering their morphologies with respect to environmental conditions, cancer cells establish a higher resonant coupling to selective signal transduction pathways of the microenvironment and damping of non-transformed cell networks' survival-dependent energy pathways (information flow). Thence, cancer cells *collapse* (select) the state function operators (measurable quantities) governing the energy dynamics of other biological observers (i.e. 'healthy' cells) in favor of their survival pathways (competitive game-strategy).

The local spatial instabilities emerging in cancer's morphogenesis can be modeled by a predator-prey system (cancer vs. healthy cells) analogous to the Turing reaction-diffusion equations:

$$\frac{\partial u}{\partial t} = S(u) - F(u, v) + D_u \nabla_u^2 u \quad \text{and} \quad \frac{\partial v}{\partial t} = NF(u, v) - M(v) + D_v \nabla_v^2 v, \text{ where } u = \text{healthy population density, } v = \text{cancer population density}$$

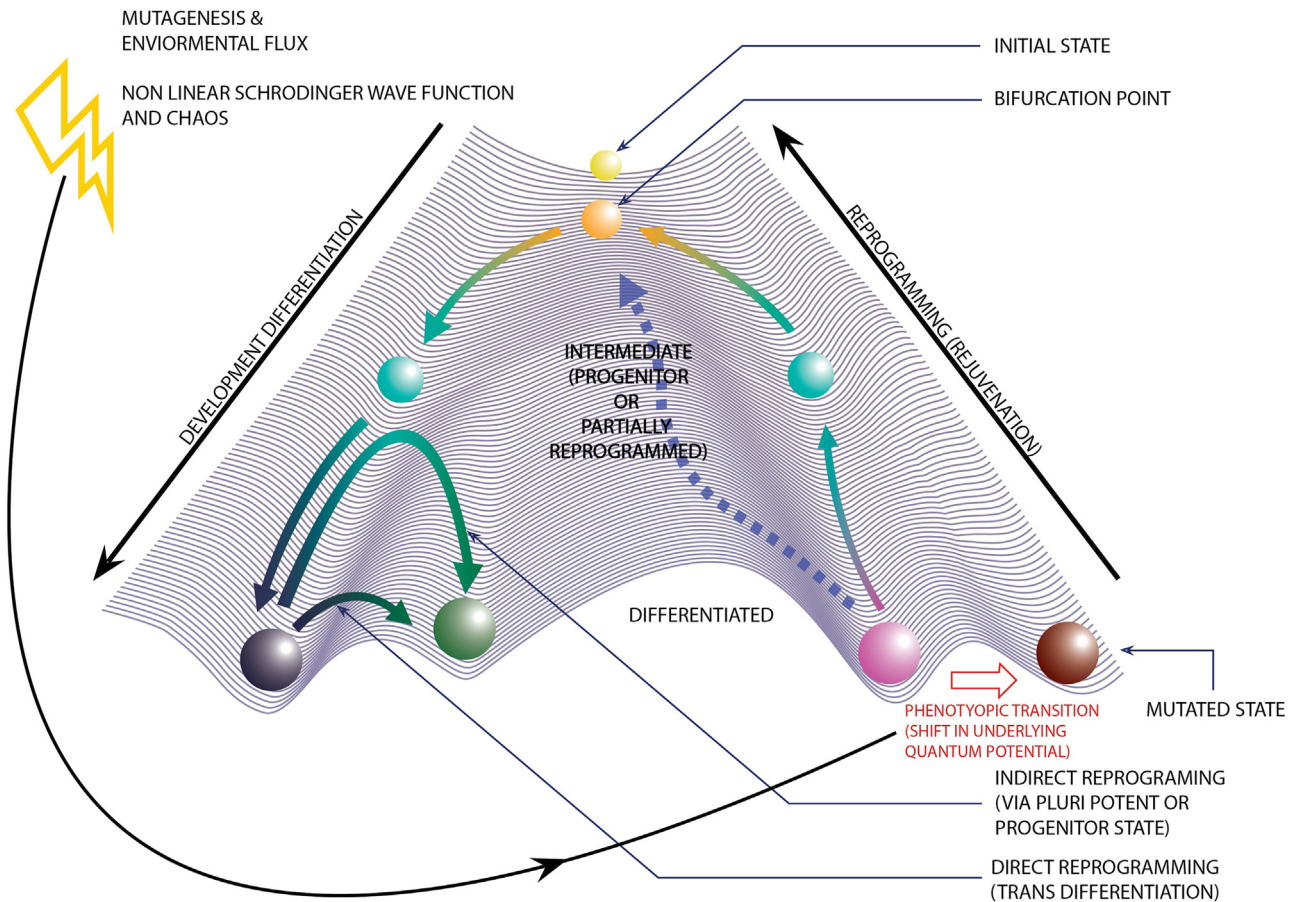


Fig. 3. The Phenotypic Landscape: A simple Waddington landscape (energy-surface) shows how cell states bifurcate from their initial conditions. The thunderbolt represents environmental perturbations and mutations, from which a curved arrow leads to the red ball indicating a mutated cell fate and the transition in its underlying quantum potential. In contrast to a classical Waddington landscape, cell state attractors can exhibit quantum wave-like properties. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lation density, S and M are intrinsic population dynamics (ex: survival rates), N =fraction of healthy cells that convert to cancer cells, F is the infiltration/invasion rate of cancer and $D_{u,v}$ are the Brownian Diffusion coefficients.

Let the subscript 1 denote cancer cells and 2 denote non-transformed cells, A the amplitude of energy-metabolome, ρ the wave-function density representing the BEC condensate (energy/epigenetic landscape), Q the quantum potential, and ω the resonant frequency of the oscillators (cells). Then: $v(t) \propto A\omega(\sin \omega t + \varphi)$ describes the velocity of the energy surface fluctuations, which for cancer cells is significantly faster: $v_1 \gg v_2$. Thereby, (1) the speed of cell division and the intensity of epigenetic flux rates (environmental information processing) is higher in cancer cells $\rightarrow \omega_1 \sqrt{A_1^2 - x^2} \gg \omega_2 \sqrt{A_2^2 - x^2}$ for a given spatial distance x and, (2) the coupling strength between cancer cells and their environment is of a greater magnitude than the coupling between healthy cells and the environment. Since $A \propto \rho$ and the Quantum potential $Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}$, $Q_1 \gg Q_2$ (Bohm, 1952a,b). This implies that cancer cells can dynamically coordinate the molecular evolution of the biosystem and its environment in pre-space (i.e. selection by the guiding Quantum potential). The BEC model allows cancer cells and their environment to be modeled as a single quantum flux system (i.e. quantum adaptive epigenetic/energy landscape).

Hence, cancer cells acquired quantum adaptive mutations to harness longevity/immortality unlike the time-bound limited healthy cells in the organism. Cancer cells' high rate of glycol-

ysis, metastatic invasion, longevity pathways and environment co-dependent changes in the expression of specific genes/receptors are not random dysfunctions but rather efficient evolutionary strategies selectively adapted by the transformed cell networks to ensure their survival (natural selection). The dynamical evolution and cell fate transitions resulting in the *observed* tumor morphogenesis can be modeled as (quantum) bifurcations on a Waddington landscape.

4. Energy landscape dynamics

As previously stated, cancer cell fates are chaotic attractors in the cells' energy (Waddington) landscape separated by epigenetic barriers with differing wave amplitudes given by Born's rule: $P = |\psi(x, t)|^2$ where the cell fate trajectories branch off as entangled fractal bifurcations. The probability waves become concentration profiles or gene/protein expression levels given experimental data obtained from epigenome-transcriptome profiling (Fig. 4). On a quantum scale, the energy landscape is a vibrating non-equilibrium spin glass lattice where cancer cell states are strange quantum attractors represented by coupled quantum harmonic oscillators. The energy landscape depicts the temporal evolution of cell states as a function of the energy potential (ex: bioelectric potential in cell-matrix coupling, epigenetic potential, Gibbs free energy in thermal-enzyme kinetics, quantum potential for quantum energy landscape, etc.) and space.

According to Levinthal's paradox, an unfolded polypeptide chain has an astronomically large # of possible conformations. If a pro-

REPROGRAMMING ENERGY LANDSCAPE

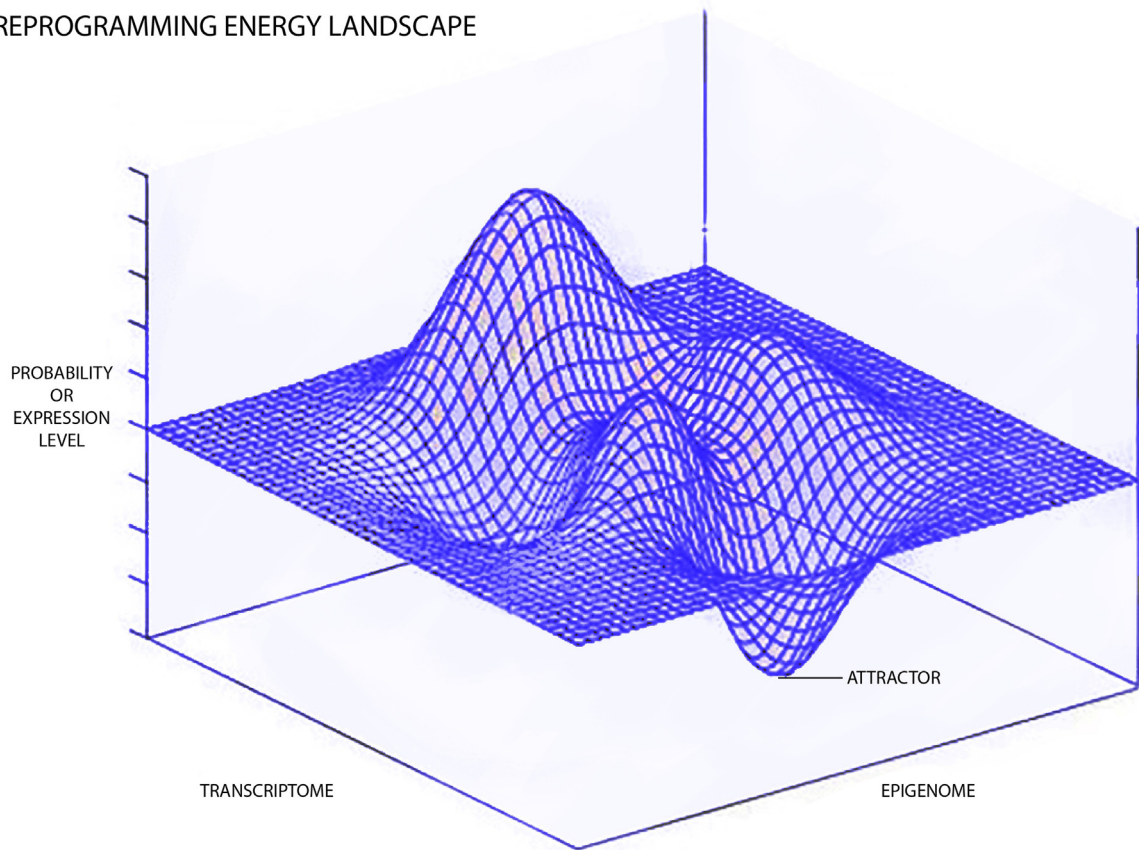


Fig. 4. The Epigenome– Transcriptome Landscape: The equivalent concept as the energy landscape in Fig. 2. The probability of the wave function becomes concentration (expression) levels with the epigenome and transcriptome profiles of cancer cells.

tein were to attain its correctly folded configuration by sequentially sampling all possible states, a time longer than the age of our universe is required to attain its native state. However, most small proteins fold spontaneously on a microsecond to millisecond time scale. The paradox can be resolved if protein-folding is sped up and guided by the rapid information flux of local interactions which then determine further folding of the polypeptide (i.e. the corresponding shifts in quantum entanglements).

Noise-induced bistable switching dynamics in such potential energy landscapes give rise to the feedback loops observed in biochemical systems and cell signaling cascades. Luo (2013) claims that conformational changes in protein folding are quantum dynamical processes and derived folding rate equations based on the quantum decoherence time of torsion states. Thence, quantum energy landscapes would portray the evolution of biopolymers as entangled density matrices of spin chains in a superposition of protein conformations interacting with a heat bath (environment-induced decoherence).

Intermediates and partially folded transitions detected in the energy funnel form a rugged surface measured by a roughness parameter (fractal dimension). Predicting bifurcations between two pathways on the Waddington landscape is highly sensitive to the epigenetic barriers, where a barrier height difference of 1 k_BT can shift the relative flux between the two potential pathways by more than a factor of 3. Adaptive mutations alter the speeds of protein folding processes, the energy resonance structures of protein conformations and the epigenetic barriers' height. Hence, by adopting a quantum energy landscape, wave dynamical equations would

provide solutions to the evolution of biological states as opposed to Boltzmann particle kinetics.

Quantum energy landscapes have been experimentally used in determining the evolution of simple biochemical systems and photosynthesis (Calhoun et al., 2009). Tunneling in the epigenetic landscape is regulated by thermal fluctuations caused by far-from equilibrium protein conformation dynamics (Onuchic et al., 2006). Klich et al. (2014) corroborated that *quantum fluctuations* produce the rugged landscape of a disordered spin glass. Quantum phase transitions on these energy landscapes are described purely in terms of quantum bifurcations arising from changes in the initial quantum entanglements of a system induced by systemic fluctuations (Zhilinskii, 2011; Hines et al., 2005). However, these models have not been used in the context of mammalian biology.

The landscape is defined by the Wigner quasi- probability path integral, where the probabilities (different peaks of the wave function) indicate the molecular concentrations and protein intensities in the cell interactome. The cell fate programming can be quantified as a dynamic and deterministic epigenetic landscape. Given the rate equations of a binary feedback loop with genes x and y , the quasi-potential difference is simply $\Delta V = -\frac{dx}{dt} \Delta x - \frac{dy}{dt} \Delta y$.

The Wigner quasi-probability distribution is the phase space analogue of quantum spin networks and can take negative values. Hence, given microarray or RNA-Seq data of the transcriptome and epigenome profiles, protein concentration and gene expressions levels can be quantified as positive or negative distribution peaks on the Wigner distribution indicating upregulation or downregulation of the screened molecules. The interference patterns of these attractors results in the observed cell phenotype.

For instance, Tumor Growth Factor (TGF- β) activation drives embryonic cells to a mesodermal fate. Whereas expressing Wnt while inhibiting Bone Morphogenetic Protein (BMP) also promotes a similar differentiation pathway. Therefore, multiple expression combinations lead to the same cell fate outcome indicating that phenotypes are determined by the quantifiable resonant energy pathways of interacting molecules. Koh (2014) provided a mathematical description of quantum spin entanglements whereby attractor states can be in a superposition of states and their wave amplitudes can be calculated in time-evolution. Hence, by adopting a quantized spin glass network to represent cancer development as a wave function would model the disordered information flow between cancer attractor states as qubits implying that the decoherence time-scales, life-times of signaling molecules, wave interference patterns, etc. can be calculated (Laflorencie, 2016; Itoi, 2016). Furthermore, in defense of the thesis that quantum fluctuations govern biological evolution and cosmic inflation, Arguin et al. (2016) extended the quantum spin glass systems to model the vacuum energy field. In theory, given the principles of reversible quantum energy pathways in the epigenetic landscape and the interconnectedness of the cancer signalling interactome (a Universal wave function), cancer phenotypes are reversible.

Frustration in Ising spin glasses account for the multi-peaked features of fitness (energy) landscapes. The entire spin glass has N spins and thereby 2^N configurations (i.e. Hilbert space). Each configuration has a Hamiltonian described as a global sum of coupled quantum spin pairs given by $H = -\sum J_{ij} (S_i \cdot S_j)$, where $S_i, S_j = \pm 1$ and J_{ij} is strength of the locally coupled spin pair. In a classical landscape, the ordering of M attractor states with respect to their metastability through some energy-like quantity $\{U_1, \dots, U_M\}$ gives a potential energy function $U(x)$ which informs the probability and direction of transitions between attractors in a noisy or perturbed system for any point x in state space. This potential is guided by an additional Quantum Potential in a quantum energy landscape.

Consider a deterministic network of N variables whose values describe the cell state $X(t) = (x_1(t), x_2(t), \dots, x_N(t))^T$. Defining a potential $U: U(x) = -\int \sum F_N(x) dX_N$, a partial notation of a quasi-potential U is given: $dX/dt = -\nabla U + F_r$ where the r denotes a term beyond the driving force deriving a non-equilibrium system. The ordering of metastable attractors is defined by quasi-potentials and is related to the classical potential V expressing the *least action path* (geodesic) that captures the epigenetic height of the landscape. Perturbations allow for attractor states to be driven to the Least Action Path of mutant phenotypes in cancer cells or stem cell state in iPSC reprogramming. The Freidlin-Wentzell potential V in discrete form for k genes is then approximated as: $V_{AB}(X(t)) = \Delta t \sum_{k=1}^{M-1} \sum_{i=1}^n \| \frac{x_i^{k+1} - x_i^k}{\Delta t} - \frac{1}{2} (f_i^{k+1} - f_i^k) \|^2$.

The classical Wentzell-Freidlin theory is used for deriving the transition rate for most probable paths connecting 2 neighboring attractors on an epigenetic landscape with small noises and Michaelis-Menten kinetics. The epigenetic state trajectory is a stochastic walk given by the rate equation: $dx/dt = F(x, \lambda, \omega) - N(x, t)$, where x is the expression level of each gene, λ is the maximum Gene expression relative to the chromatin site (epigenetic barrier), ω is the genetic oscillator's frequency and N is the noise term. The noise term is expressed in Brownian motion as: $\langle N_i(t) N_j(t') \rangle = 2D_{ij}(t-t')$, where D characterizes the Diffusion Coefficient. The quasi-potential of the landscape is: $V(S) = -\ln P(S)$ for a non-equilibrium system with driving force S and probability distribution P .

In attractor theory, noise causes totipotent/pluripotent cells to fluctuate back and forth in the basin of attraction leading to heterogeneity and priming from a stem cell fate to differentiated states while the fluctuations of the environment is identical to the vacuum model or a thermal heat bath. Energy fluctuations are the physical counterpart of mutations, whereas the search for hyper-

stability is the role of evolutionary selection. Wu et al. (2016) developed a classical potential flux landscape to describe spatially inhomogeneous thermodynamic systems via Langevin and Fokker-Planck equations. The flux landscape produces a Mexican-Hat like topology when fluctuations are small. With a multi-particle system under constant unpredictable noise following the non-equilibrium fluctuation-dissipation theorem (FDT), the Mexican-hat evolves to a rapidly fluctuant vacuum foam-like structure where the barrier height separating oscillators (attractors) measures the cell state robustness. It must be noted that FDT gives rise to Gauge theories from which the entire Standard Model is constructed.

Wu et al. (2016) verified that the above defined Mexican-hat landscape models the cell-cycle potential and cell dynamics. The Fokker-Planck dynamics is given by: $\frac{\partial}{\partial t} P_t(\vec{q}) = -\nabla \cdot (\vec{F}(\vec{q}) P_t(\vec{q})) + \nabla \cdot \nabla (D(\vec{q}) P_t(\vec{q}))$ where the probability flux is $\vec{J}(\vec{q}) = \vec{F}(\vec{q}) P_t(\vec{q}) - D(\vec{q}) \nabla P_t(\vec{q})$. Then the thermodynamics of a spatially inhomogeneous system on Hilbert state space is $\partial_t P_t[\phi] = -\int dx \cdot \delta_{\phi(x)} \cdot \vec{J}(x, t)[\phi]$, where the potential landscape is $S[\phi] = -\ln P_t[\phi]$ and the transient flux velocity field is $\vec{v}(x)[\phi] = \frac{\vec{J}(x, t)[\phi]}{P_t[\phi]}$. Hereby, I further extend this model under a Bohmian interpretation such that an underlying Quantum potential selectively adapts the evolution of the epigenetic landscape (Fig. 5).

The Navier-Stokes equations describe the turbulent dynamics of the potential energy landscape (Wang et al., 2008). Consider the epigenetic landscape as a system with the total energy defined as $E_{tot} = \frac{1}{2} \int |\nu(x)|^2 dx$ where $\nu(x)$ is the velocity field. The energy flux cascade $T(k)$ for the potential landscape is $\varphi[\nu]: T(k) = R \{ \langle \nu^*(k) D^s(k) \cdot \nabla_{\nu(-k)} \varphi[\nu] \rangle_s \}$ and $T(r) =$

$\frac{1}{4} \nabla_r \cdot \langle |\delta \nu(x, r)|^2 \delta \nu(x, r) \rangle_s$ for inhomogeneous velocity fields $\therefore$ the energy dissipation rate with respect to the energy flux is $\varepsilon = -(\frac{1}{2\pi})^3 \int T(k) dk$. The energy turbulences follow the Kolmogorov-Richardson energy cascades.

Reprogramming cell fate involves rebooting the genome via reconfiguring the genetic networks (ex: introducing transcription factors or exogenous proteins) and erasing the covalent chromatin marks, which at the level of individual cells has a frequency $< 1\%$. Thus, a critical frequency of the mutated genetic network oscillations (epigenetic barriers) must be overcome to adopt transformed cell states. Since somatic cell phenotypes are inter-convertible (epigenetic modifications are reversible), by knowing the energy density of the epigenetic barriers corresponding to the perturbed signaling cascades from its default state, tumor cell phenotypes can be reverted. Transdifferentiation between neighboring attractor branches in the developmental landscape can be attained by the over expression of a single Transcription Factor (TF). Thereby, the reprogramming dendrogram is obtained by comparing the transcriptome-epigenome landscapes of cancer cells and their healthy (untransformed) counterparts.

Each cell fate/type is a biological observable (Hermitian operator on Hilbert space or a state vector/trajectory on Phase space) that could be represented as a spin glass configuration. For instance, the evolution of the state vector $q(x, t)$ describing the trajectory of the concentration profiles of the different reagents in morphogenesis can be simplified by the diffusion equation: $\partial q = D \Delta q + R(q)$. A trajectory corresponds to a specific spin glass time crystal (condensate) (Edwards and Anderson, 1975).

Hopfield (1982) defined memory/neural networks as spin glasses and Bryngelson et al. (1995) applied them to protein folding. Turing-Hopf bifurcations can explain the chaotic self-organization of energy dissipative systems and the morphogenesis patterns produced by biological oscillators (Baurmann et al., 2007). The coupling interactions of the biological oscillators result in morpho-

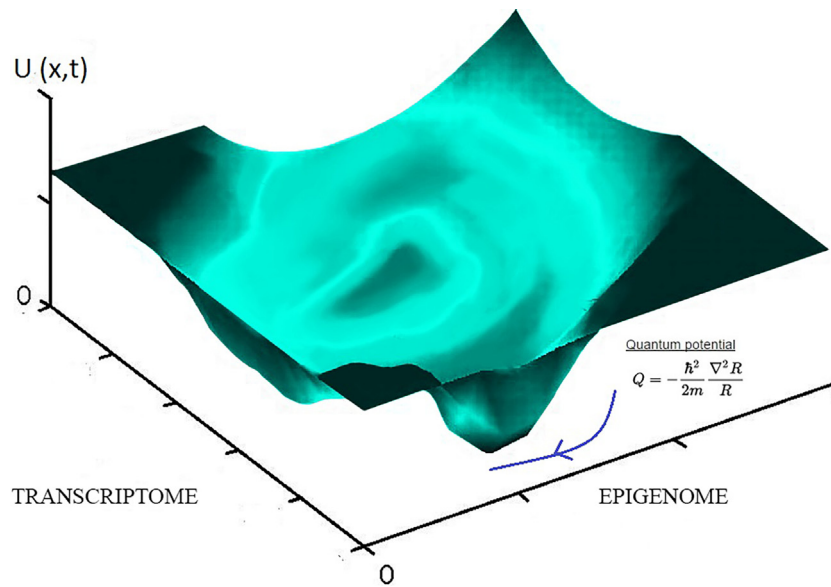


Fig. 5. Epigenetic Flux Landscape: The transcriptome–epigenome landscape is shown as a Mexican-hat like non-equilibrium topology with minimal noise driven by a nonlinear pilot wave field denoted by the curved arrow with the Bohm's Quantum potential equation (i.e. adaptive selection in pre-space).

logical patterns if a symmetry-breaking bifurcation (perturbation in transition pathways) occurs at their critical threshold frequencies. At the sub-critical threshold, dissipative solitons are observed (Strogatz and Stewart, 1993).

BECs model biological condensates (Wu and Austin, 1978; Tuszynski et al., 1984). Davydov established that the vibrational energy of proteins, nonlinear dynamics of DNA, H-bond couplings and the spatial propagation of energy in biological systems are described as self-interacting solitons. Cancer (stem) cells and micrometastases can be modeled as such dissipative structures whereby nonlinear lattices of coupled solitons form BEC condensates (Christiansen and Scott, 1990; Forner, 1997). Thereby, BECs model the coherent vibrations of cell divisions, protein synthesis and nonlinear macromolecular dynamics in the millimeter range (Bhaumik et al., 1976; Scott, 1982).

Biswas and Pacheco (2003) described cancer growths as dissipative Bose-Einstein condensates following the Gross-Pitaewskii Equation (GPE) based on pre-vascular carcinoma cultures. With vasculature, this becomes a hyper-chaotic condensate where angiogenesis produces fractal bifurcation structures. The wave interference patterns of tumor modeling BECs are given by the Inverse Scattering Transforms of a Nonlinear Schrödinger equation (GPE) (Strecker et al., 2002). The cell states would then behave as waves in a superposition of bifurcating trajectories in their epigenetic landscapes. For a thermodynamic state (i.e. BEC) the configuration probability can be simplified as: $P(J_{ij}) = e^{-BH}/ZB$, where Z is the Partition function and $B = (k_B T)^{-1}$. Given the time-energy uncertainty of a quantum oscillator with a frequency ω , the spin probability is $P_t(S_x = \pm \frac{\hbar}{2}) = \left\{ \cos^2\left(\frac{\omega t}{2}\right); \sin^2\left(\frac{\omega t}{2}\right) \right\}$ (Montanari and Mezard, 2009).

Cancer cell states are evolutionary by-products of chaos (i.e. exponential hypersensitive dependence on initial conditions). Their nonlinear dynamics are best-modeled as strange attractors (fractals). The Poincare section of a metastatic cancer model will exhibit strange attractor dynamics implying that cancer topologies can be differentiated from non-tumors by their fractal behavior (Li and Ma, 2016). A simple strange attractor, the Lorenz attractor:
$$\begin{cases} \dot{x} = \alpha(y - x) \\ \dot{y} = (c - \alpha)x - xz + cy \\ \dot{z} = xy - \beta z \end{cases}$$
 models time delays in cell com-

munication signaling feedback loops and the reaction-diffusion of metastasis and tumor growth. Therefore, cancer cells are hyper-chaotic attractor systems with dynamical nonlinear couplings (i.e. the rates of quantum entanglements and quantum decoherence are turbulent). The epigenetic state transitions in the tumor development landscape correspond to *quantum chaotic attractors*. The phase-portraits are determined by the maximal Lyapunov exponents and can compute tumor expansion or spreading rates (Sparrow, 1982; Kadji et al., 2007).

Lang et al. (2014) used epigenetic landscapes to explain partially reprogrammed cell fates with a non-Lyapunov pseudo-potential. Partially reprogrammed cells are spurious attractors (hybrids with smaller basins of attraction). Each Transcription Factor (TF) can be in a state $S_i = +/ - 1$ (active or inactive) or a superposition of states. Consider asynchronous updates in stable attractors (genetic networks) following Glauber dynamics. Given $B = 1/T$ as the effective noise parameter and h_i as a local field (landscape gradient), at each unpredictable update a TF is then selected by the probability: $P(S_i(t+1)) = e^{Bh_i(t)S_i} / (e^{Bh_i(t)} + e^{-Bh_i(t)})$.

However, epigenetic processes and cancer cells states are unstable strange attractors with fluctuant energy dissipation (Huang et al., 2009; Li et al., 2016). The cancer metabolome is a highly turbulent system where the hyperchaotic attractors and repellers are strange fractals (Bizzarri et al., 2011). Letellier et al. (2013) further demonstrated that the bifurcation diagrams of tumor growth correspond to chaotic logistic maps. In the proposed model, the unstable cell attractor may circulate at different levels around the walls of the landscape's depression, exhibit wave dynamics and quantum tunnel through epigenetic barriers (quantum oscillatory states) as opposed to the classical view of fixed stationary states during transitions. Tumor growth and tumor vasculature have been validated as fractal topologies emerging from chaotic attractor dynamics (Sedivy and Mader, 1997; Gazit et al., 1997; Dokukin et al., 2015).

The fluid dynamics of the quantum energy landscape is given by quantum Navier-Stokes equations. $P_{1 \rightarrow 2}(t) = |\psi(x, t)|^2 = \left(\frac{2\pi}{\hbar} |H|^2 \cdot \frac{1}{\delta \varepsilon} \right) t$ calculates the quantum transition density, where the expression in the parentheses is the transition rate given the energy flux ε . The energy density of states is $\rho(E) = \frac{1}{\delta \varepsilon}$. Given two

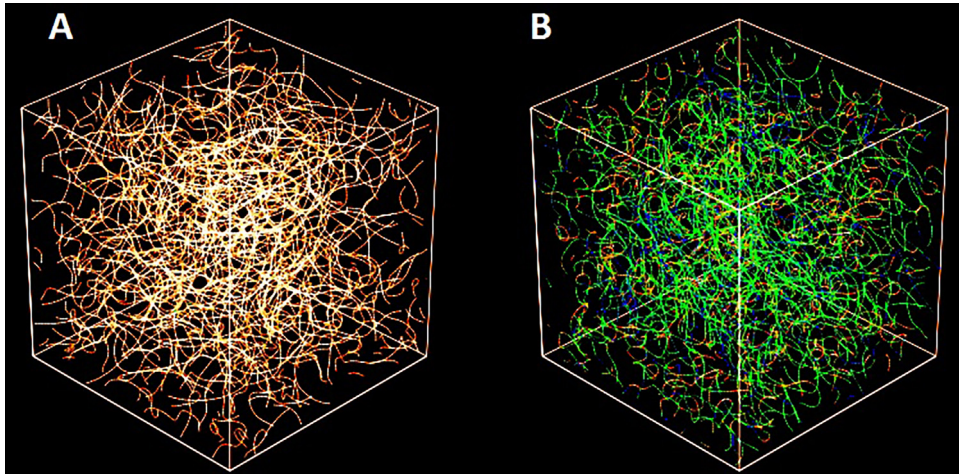


Fig. 6. Quantum Turbulence: (A) The quantum vortex tangles and filaments of a Bose-Einstein Condensate's energy cascades are shown. (B) The energy fluctuations of a quantum vacuum field demonstrate equivalent topologies as the BEC vortex tangles implying the BEC model could simultaneously define the energy landscape dynamics (i.e. the evolution) of biosystems (epigenetic landscape) and cosmic inflation.

state bifurcations 1 and 2: $H = \langle \hat{H} | 2 \rangle = \int_{-\infty}^{\infty} dx \psi_1^*(x, t) \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_1(x) + V_2(x) + g(x, t) + Q(x) \right] \psi_2(x, t)$. Given the quantum energy landscape is a nonlinear wave function, it is then best described by the Gross-Pitaevskii Equation of a BEC condensate where $g = \frac{4\pi\hbar^2 a}{m}$ is the vortex coupling parameter given the scattering length a .

It must be emphasized that the active principle of information flow in biological systems is energy, which under the AdS/CFT correspondence (quantum gravity models) and SVT (superfluid vacuum theory) is characterized as a holographic quantum turbulent superfluid-BEC that nonlinearly fluctuates in hyperspace while obeying the wave-particle duality (Comper et al., 2012; Lepe and Peña, 2016). Therefore, in principle, the hydrodynamical equations of BEC quantum turbulences discussed in the context of epigenetic condensates and Orch OR theory (cytoskeletal networks) must equally apply to the topological dynamics of quantum cosmology implying space-time can be represented as quantum vortex tangles (Fig. 6). Moreover, Couder et al. (2014) corroborated that deterministic quantum fluid dynamics can experimentally validate Bohm's pilot wave theory at the millimetric scale.

Bose-Einstein Condensates (BECs) are macroscopic spin glass networks modeling quantum turbulence (Aspelmeier and Moore, 2004; Tsubota and Kasamatsu, 2012). Protein folding funnels and epigenetic landscapes can be described as hysteretic BECs following the Gross-Pitaevskii equation (Mueller, 2002). Molecular transport along the DNA and of biological molecules can be modeled as BECs (Denschlag et al., 2000). As previously stated, weakly interacting Fröhlich condensates apply to many biological structure dynamics (Wu and Austin, 1982). The DNA has been shown to resemble a BEC during its melting phase and denaturation (Porter et al., 2005; Neher and Gerland, 2008). Moreover, many mathematical models exhibit the DNA as topological solitons, an agitated or interactive topology of which may results in a BEC (Wüster et al., 2009).

In theory, quantum spin lattices of BEC condensates model cancer cells and their morphogenesis as quantum coherent multi-particle densities. Despite their separate local dynamics appearing random-like due to chaos, the densities are interconnected into a single global system. The energy flux of the cancer environment (heat bath) can be mapped by the same equations as the energy fluctuations of the quantum foam model (Scandurra, 2001; Carleo et al., 2009). By adopting a quantum fluid-BEC model, the

tumor-environment dynamics and thereby cancer evolution can be depicted as a singular quantum turbulent flux system (Fig. 7).

As mentioned, measuring the quantum decoherence rate (the nonlinear decay of the off-diagonal elements of a density matrix) is subjected to the uncertainty principle in the probabilistic interpretation of QM where only average estimates of the decoherence rates are possible. However, by defining cancer epigenetic landscapes as a macro quantum coherent system, the Bohmian GPE equation is a suitable candidate for solving the Nakajima-Zwanzig equation, where the off-diagonal terms are described as the quantum coherence of dissipative fluxes (i.e. there is no differentiation from a fluctuating term and observed/coherent term since the *entire system is a fluctuation*).

Consider a 2-level atom (Rabi oscillator) coupled to a heat bath (collection of harmonic oscillators) following the Jaynes-Cummings model to derive the Nakajima-Zwanzig master equation for the system's density matrix. The Hamiltonian of the entire system is $H = \hat{H}_{field} + \hat{H}_{subsystem} + H_{interaction}$. Note that the Hamiltonian formulation is identical to the energy operator of a biosystem described by Davydov's solitons. Thus, the The Hamiltonian of the heat bath is $\hat{H}_{field} = \sum_k \omega_k b_k^+ b_k$, where b_k^+ and b_k are the cre-

ation and annihilation operators obeying bosonic commutation. $H_{subsystem} = \omega_0 \sigma + \sigma$, where ω_0 = transition frequency and $\sigma_+ = |10\rangle$ and $\sigma_- = |01\rangle$. The interaction Hamiltonian derives a Lorentzian spectral density given by $H_{int} = \sum_k (g_k \sigma_+ \otimes b_k + g_k \sigma_- \otimes b_k^+)$.

However, this equation must be treated as a nonlinear Nakajima-Zwanzig equation to portray the complexity of non-equilibrium biological systems. Given the resonant frequency of field vibrations (ex: cytoskeletal networks, epigenetic landscape) the Hamiltonian of the environmental flux field (turbulent heat bath- chain of bosonic nonlinear oscillators) replaces the $\hat{H}_{field}$ and is given as: $\hat{H}_{environment} = \frac{m\omega^2}{2} (\hat{x}^2 - (\frac{\hat{p}}{m\omega})^2)$. $\hat{H}_{biosystem}$, the Hamiltonian of the subsystem (cancer cells coupled to the environment) can be calculated as the Hamiltonian operator of a time-evolving multi-particle system represented by a Bohmian GPE function ψ_N such that $\hat{H}_{biosystem}(\psi_N) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(x_1, \dots, x_N, t) + Q(x, t) + g|\psi(x_1, \dots, x_N, t)|^2 \right) \psi_N$.

The interaction term is the epimetabolic flux energy given by: $E(A, B) - (E(A) + E(B))$, where $A = H_{field}$ and $B = H_{sub}$. Given the entire flux system is a thermodynamic BEC condensate, the

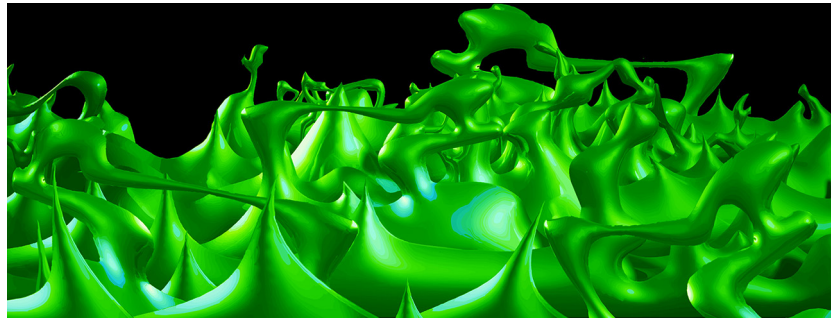


Fig. 7. The Quantum Foam: The quantum space-time topologies discussed (superfluid vacuum). As proposed, the quantum foam is a dynamical spin glass with energy turbulence identical to those observed in the epigenetic landscapes of cancer cells.

$H_{interaction} = \varepsilon_{epimetabolome} = \left[\frac{g}{2} |\psi_N|^4 + k_B T + V_{ext} |\psi_N|^2 + Q(x, t) \right]$ where g is the quantum vortex coupling parameter, k_B = Boltzmann constant, V_{ext} = bioelectric potential, T = system temperature and Q is Bohm's quantum potential. The interaction Hamiltonian describes the coupling between cancer cells and their environment (including non-transformed cells). Furthermore, the cancer-environment co-evolution is mapped as a singular quantum dynamical flux system (i.e. quantum adaptive epigenetic/energy landscape).

5. Bioelectromagnetism, endopsychedelics and water networks

Many studies corroborate that a major regulator of environmental noise in biological systems are water molecules (cytoplasm) and their H-bond networks in protein folding and biopolymer/solute dynamics. According to Pokorny and Kobilkova (2013), water ordering coordinates the electrical field induced microtubule Fröhlich condensation (~8 MHz) fueled by the quantum heat bath of mitochondrial networks. However, he defines cancer transformation as a mitochondrial dysfunction and reduction of the coherent energy state of cell networks. In contrast, I propose the non-equilibrium fluid-like morphogenesis of cancer as a *quantum adaptive selection* for enhanced survival in its fluctuant environment. Cancer diagnosis is based on frequency selective absorption of EM waves by malignant tumors due to their lower water ordering (Pokorny et al., 2011). Moreover, the emission of mitochondrial biophotons have been claimed to form quantum effects with microtubules (Rahnama et al., 2011).

Water molecules control protein motion and folding (Qin et al., 2016). Using Femtosecond Fluorescence Emission Spectroscopy, protein folding was shown to be executed by surrounding water molecules near the picoseconds range. Water molecules formed hydration shells coordinating the protein fluctuations. Although protein conformations are amino acid sequence dependent, the final configuration was shown to depend strongly on water fluctuations. Moreover, Siebert et al. (2015) using 2-D infrared spectroscopy demonstrated that water regulates vibrations within DNA topology.

Montagnier et al. (2010) further claims the wave function collapses for biological macromolecules such as DNA occurs due to the interaction of background electromagnetism and water. Hence, DNA transduction is a QFT process. The water H-bond networks are proposed to orchestrate the transmission of DNA waves. He claims water can store the Extremely Low Frequency (ELF) waves emitted by the DNA into aqueous nanostructures. Post-serial dilutions, the imprinted information waves are retained and the DNA can be amplified (reconstructed) by Polymerase Chain Reaction

(PCR) implying there is a link between bioelectricity and coherent biopolymer (matter) waves (Montagnier et al., 2015).

Thereby, the DNA-water system emits EM signals and requires a background EM-ELF near the Schumann resonances (starting at 7.83 Hz) to form nucleotide polymerization. Low frequency biological oscillators such as Davydov's soliton model, Fröhlich coherence and Orch OR theory have been proposed. In support of the DNA waves transduction hypothesis, quantum mathematical models describing the electric dipole mediated DNA-enzyme kinetics in water environment have been formulated. Kurian et al. (2016) further propose DNA waves occur from spontaneously broken gauge symmetries.

In a quantum metabolome, mitochondrial networks are coupled harmonic oscillators that capture and transmit energy from oscillating electric fields. The proton-motive force across the quantum mitochondrial membrane is simplified as: $\Delta p = \Delta \psi - (2.3RT/F) \Delta pH$, where $\Delta \psi$ is the membrane potential, F is Faraday's constant and the pH should be acidic in most cancer cell environments (L.A. Demetrius et al., 2010). The Goldman-Hodgkin-Katz equation can calculate the approx. membrane potential of cells amidst electrochemical gradients. The electrical stimulation of human skin increases angiogenesis and wound healing (Ud-Din et al., 2015). The EM field generated by chromatin oscillation is the driving force for chromosome packing (conformation) during M phase (Zhao and Zhan, 2012).

It is well-established that changes in the electric field of cells are markers of cytoskeletal-matrix rearrangements and tissue morphogenesis. When a cell loses some of its negative charge it helps tumors spread. Although membrane potentials are weaker, cancer cells are conjectured to have stronger electric field couplings to their environmental tissues, permitting their membrane potentials to dynamically fluctuate depending on the stage of tumor progression and environmental conditions. It must be noted that varying dynamics exist for the *endo*-membranes (ex: mitochondrial membrane vs. plasma membrane, etc.).

The internal (fluid) electric fields are significantly stronger in cancer cells than healthy cells. Kopelman et al. (2006) used nanovoltmeters and nano-scale encapsulated voltage sensitive dyes (PEBBLE technique) to measure electric fields as strong as 15 million volts per meter in glioma cells' internal fluid (5X stronger than a lightning bolt). While the cytoplasmic potential is significantly high, the plasma membrane potentials of cancer pathologies and rapidly dividing regenerative cells (stem cells) are weaker than healthy cells (Yang and Brackenbury, 2014). The weak membrane potential and symmetric division of (cancer) stem cells permits cytoskeletal flexibility, regenerative potency and fluid-differentiation capacities whereby cells selectively adapt their morphology to environmental changes.

Levin (2011) propose that by altering this *bioelectric code*, cell-communication signals can be hijacked. They demonstrated how

the endogenous voltage potential controls cell patterning, embryogenesis, regeneration and morphogenesis in cancer cells. Electric fields can affect gene expression and epigenetics implying a top-down causation effect. Thus, limb regeneration, wound healing modulation, selective tissue patterning and tumor morphology reprogramming are possible through manipulating these bioelectromagnetic networks.

Transmembrane voltage level is the initiator of dispersed metastases and tumor morphogenesis. For instance, KCNE1 misexpression induces hyperproliferative-invasive cell behavior. Metastatic potential correlates with enzymatic degradation of basement membrane collagen. The cell lines with the highest incidence of spontaneous metastasis exhibit the greatest level of type IV collagen-degrading activity. Depolarization of resting potential by voltage sensitive fluorescent dyes can detect cancer *in vivo*. Oncogene-induced tumorigenesis requires depolarization, and artificial hyperpolarization can suppress this effect, despite high levels of oncogene protein, through a butyrate and histone deacetylase mechanism.

A hyperpolarized V_m is required during stem cell maturation, while depolarization maintains pluripotency/regeneration. V_m depolarization may maintain cells in an undifferentiated stage at the gene expression level. The depolarized V_m helps maintain a population of undifferentiated CSCs. Rapidly proliferating cancer cells and proliferating somatic cells require a depolarized plasma membrane potential. Also the membrane potential V_m changes in response to the cell cycle and the efflux/influx of potassium, sodium, etc. ions

Given that the fluctuation of V_m can functionally regulate tumorigenesis, differentiation, and promote cancer progression, it may serve as a potential marker for tumor detection and treatment, and prognostic indicator to monitor cancer progression. A depolarized V_{mem} is a characteristic of induced tumor-like structures (ITLS) in *Xenopus* models (bioelectric signatures of pathogenesis). Butyrate (HDAC inhibitor) may suppress ITLS by hyperpolarization. Hence, selectively controlling V_{mem} may reprogram tumor morphogenesis. For instance, Verapamil (a voltage-gated Ca^{2+} channel blocker; clinical hypertensive) can induce cell death in breast cancer cells since their membrane potential is depolarized. Voltage-gated sodium channels (VGSCs) facilitate and regulate cancer invasion/metastasis. VGSC blocker inhibitors, such as phenytoin and flunarizine are anti-metastatic targets.

However, membrane potentials must be evaluated in other internal membranes in cancer cells. Formation of the mitochondrial membrane potential depends on the flux of respiratory substrates, enzymes, protons/electrons, ATP/ADP and Pi through voltage-dependent anion channels (VDAC). Since tubulin promotes single channel closure of VDAC, tubulin is hypothesized as a dynamic regulator of mitochondrial membrane potential. The inhibition of VDAC by free tubulin limits mitochondrial metabolism in cancer cells.

The mammalian membrane voltage evolution exhibit bistability and non-heritable memory transmission like the epigenetic landscape. Endogenous biochemical networks store non-genetic patterning information during developmental regeneration. Endogenous bioelectric fields mediate wound healing by coordinating epithelial closure, activating ECM proteins, initiate appendage regeneration, control differentiation of mammalian adult and embryonic stem cells. Thereby, bioelectric networks function as *epigenetic controls* of growth, form and regeneration.

Somatic cells can maintain a distinct resting potential as a state memory. When expressing different ion channel subsets, bistable memory formation occurs. The nervous system implements memory and information via electrical communication in cells. Likewise, electric fields generate non-genetic patterning information in non-excitable cells but usually on a slower

timescale. Ion channels are gated post-translationally. Thus, even when the proteome/transcriptome is unperturbed, protein phosphorylation and epigenetic PTMs can open/close ion channels (i.e. from cardiac-neural cells to extracellular matrices, all tissues constantly exchange information via bioelectric networks). Hence, somatic cells can write, erase and rewrite discrete voltage states as cell memory (analogous to epigenetic inheritance) and can occupy many different stable levels of V_{mem} (entanglement of bioelectric attractors).

Lobikin et al. (2012) further propose that bioelectrical signaling among non-excitable cells coupled by the gap-junctions stimulates neural network like dynamics. Gap junctions contribute to the synchronous firing of electrical synapses and regulate tissue morphogenesis. They connect cells and mediate electrical coupling with the ECM. The Gap junctional communication can be modified by GPCRs and post-translational/covalent modifications through enzymes such as kinases. In experimentation, the Hodgkin-Huxley model's membrane potential yields a large range of thresholds with simultaneous firings as opposed to the classically predicted narrow window (Naundorf et al., 2006). Hameroff suggested that this temporal variability can be explained by the Heisenberg uncertainty principle and thereby the decoherence (self-gravitation) energy of microtubules accounts to $E = \hbar/t$.

As mentioned, Fröhlich (1968) first applied BECs to model the quantum coherence of neural networks oscillating in a heat bath. Craddock et al. (2012) suggested that neural memory is encoded in microtubule lattices by CaMKII phosphorylation (Calcium/calmodulin-dependent protein kinase II) implying the synaptic (electrical) plasticity of cytoskeletons coordinate information storage. Tubulin qubit lattices and microtubule networks form the electrical nervous system of cells where the information energy density flux defines consciousness (Hameroff and Penrose, 2014). Each tubulin consists of a multiple quantum states due to epigenetic states/PTMs and Covalent modifications. Thereby, fractal sub-dimensional networks formed by microtubules and MAPs (microtubule associated proteins) guide synaptic transmission and calcium induced CaMK2-microtubule phosphorylation enables long term quantum memory potentiation. In Alzheimer's disease, neurofibrillary tangles of Tau proteins form through hyperphosphorylation and microtubules disintegrate to form amyloid plaques. Tau also acts as a traffic signal for motor proteins in synaptic transport. Hence, Orch OR theory conjectures a quantum-epigenetic explanation to ageing-related diseases.

Fisher (2015) substantiated an alternate neural quantum memory model where quantum information processing occurs through nuclear spins in the brain. Phosphorus is suggested to serve as a neural qubit, where the Posner molecule $Ca_9(PO_4)_6$ protects the qubits through quantum coherence. Hence, Posner molecules regulate the inter-balance between quantum entanglements and quantum decoherence dynamics. This implies epigenetic mechanisms such as phosphorylation may determine the quantum decoherence rates and entanglement strengths of epigenetic/biological signaling processes.

Moreover, Hameroff proposes that tubulin vibrations can increase their frequency range by the action of anesthetics and psychedelics (Hameroff and Penrose, 2014). Tubulins are proteomic active sites for the action of general anesthetics (Craddock et al., 2015). The fractal neurons allow quantum vibrations of MTs to increase from MHz to GHz through the action of psychedelics (i.e. quantum energy (electron) resonance). Hence, distorted time perception and altered states of consciousness (brain wave patterns) in the context of quantum coherence and quantum wave effects must be further investigated.

In the neurological system, the pineal gland interconnects light to spacetime through the circadian clock. The ticking frequencies of the circadian clock are regulated by PTMs. Further

insight into biochemical pathways associated with the pineal gland (ex: endocrine signals, tryptamines, arachidonic acid metabolism, endocannabinoid systems, etc.) must be scrutinized within cancer signaling to define *consciousness* (Guindon and Hohmann, 2011). In addition, the effects of endopsychedelics molecules such as dimethyltryptamine (DMT), pinoline, oleamide, virodhamine and anandamide (AEA) known to alter our states of consciousness, must be explored in context of cancer epigenetics. eCB signaling is essential in the implantation of embryo (ERK/Ca²⁺/AEA pathway), anti-cancer pathways (e.g. the ceramide pathway- Erk/JNK/p38; Akt-PI3K pathway, PKA-cAMP-GPCRs, etc.), anti-neurological pathologies and depression. Furthermore, investigating the endocannabinoids (eCB)- inflammatory link to cytoskeletal morphogenesis could provide a neuro-immune link to carcinogenesis. As cancer cells are byproducts of altered cytoskeletal dynamics, studying the signaling cascades corresponding to its inflammatory changes may provide a neurochemical link to epigenetic inheritance through the endocannabinoid pathways (e.g. Is there a link between the eCB-DMT pathways through tryptophan residue binding on microtubules in calmodulin-dependent Tau-MAPs phosphorylation (epigenetic) interactions?). Hence, eCBs and serotonin-like endopsychedelics (e.g. DMT) may interconnect the mental/cognitive state of a conscious organism to its inflammatory-immune pathways resulting in cancer.

Tumors actively interact with their microenvironment to maintain malignant phenotypes and one of such signaling cascades is the Tryptophan pathway. For instance, the pineal gland's melatonin epigenetically regulates biological clocks (Korkmaz et al., 2009). The TRP (tryptophan) metabolism derives melatonin from the serotonin pathway through HIOMTs (acetylserotonin-O-methyltransferases) and SNAT (serotonin N-acetyltransferases) along with cofactors such as SAM/SAH implying an epigenetic control of the *endo*-hallucinogen's biosynthesis.

Melatonin is a key regulator of carcinogenesis. It promotes circadian rhythm-induced proliferation of cells through the interaction of CLOCK/HDAC3/c-Myc and inhibits cancer metastasis. The circadian clocks of cancer cells are altered intrinsic space-time tracking systems coupled to cell cycle dynamics, which are used to anticipate environmental changes and adapt to appropriate times and conditions for enhanced survival (especially in a resource-fluctuant environment). The cancer- stroma/ECM circadian rhythms regulate the dynamic chromatin transitions. Thereby, selective plasticity and epigenetic adaptation to nutritional/metabolic fluctuations are enabled.

Meanwhile, at the molecular level, CLOCK is an example of a transcription factor involved in the ticking frequencies of cellular clocks that also serves as a Histone acetyl transferase (HAT). It is linked to signaling pathways such as SIRT1/BMAL which are associated directly to ageing and cancer. The Clock-Controlled Genes (CCGs) and associated epigenetic changes in cancers have been well established (Feillet et al., 2015). The clock cycle of brain EEG waves behave as quantum harmonic oscillator couplings implying it may conform a BEC landscape.

In contrast to the well-established tumorigenic effects of ionizing radiation, exposure to ELF has been hypothesized to decrease melatonin production and promote cancers (Touitou and Selmaoui, 2012). MCF-7 cells exposed to ELF altered gene expression of estrogen receptors cofactors that may contribute to in vitro tamoxifen resistance (Girgert et al., 2009). Nocturnal light exposure depresses the conversion of serotonin to melatonin in the pineal gland. Chronic exposure of rats to 60 Hz electric field were shown to severely inhibit night-time melatonin production via disturbance of the hypothalamic SCN (suprachiasmatic nucleus) through indole-epigenetic cascades involving HIOMTs, NATs and pineal/blood melatonin concentration changes (Reiter, 1993).

Gomes et al. (2014) identified N, N-DMT in the supernatant of cultured SK-Mel-147 cells. (74) They established that when Dimethyltryptamine (a potent endogenous hallucinogen) is added to cancer cell cultures, it increases peroxidase activity. Therefore, the methyl/acetyl switches of endogenous psychedelics such as DMT and serotonin-melatonin pathways will interlink the neuro-epigenetics of consciousness to cancer epimetabolics.

Furthermore, Sigma-1 Receptor is a regulator of cancer cell electrical plasticity. It shapes cancer cell electrical signatures in response to environmental conditions implying an epigenetic target in cancer. It can abrogate invasive functions of ion channels in cancer morphogenesis. Sigma-1 receptors are found in high densities in many human cancers. It is also located at the mitochondria-associated endoplasmic reticulum membrane (MAM) regulating calcium signaling and cell fate control. It is a major link between bioelectricity and morphogenesis in cancer cells through gap junctions. These channels also influence the cell cycle and biological clocks. DMT is a ligand for Sigma-1 receptors, implying perhaps the bioelectricity of cancer cells might be linked to the pineal gland and its neurological clock mechanisms.

High [] of Sig 1R ligands (ex: DMT) inhibits small conductance K⁺ channel, Na_v1.5, K_v1.4, voltage gated Ca²⁺ channels, acid sensing ion channels and volume gated Cl⁻ channels. DMT inhibited hNav1.5 in HEK293 cells and was shown to regulate the bio-electric fields of cancer cells (Fontanilla et al., 2009). Membrane proteins alone can regulate the coupling between sigma receptors and K⁺ channels implying that matrix-membrane structures are energy transduction modulators. Craddock et al. (2012) showed that aromatic amino acid rings (tryptophan, phenylalanine and tyrosine) are densely arranged in non-polar hydrophobic regions and behave as quantum channels in tubulins. Thereby, the binding of psychedelics causes quantum vibrational frequency shifts (altered state of consciousness) through quantum resonance.

The interaction of excitatory and inhibitory neurons distributed within a 2D cortical sheet can be modeled by Wilson-Cowan equations (1972), where spatially periodic patterns induced by hallucinogens (ex: stripes) can bifurcate from homogeneous low activity state as Turing instabilities. Such morphogenetic structures are governed by the activator-inhibitor dynamics of the chemical morphogenesis equations derived by Turing (1952). The Ermentrout-Cowan model (1979) of visual hallucinations further demonstrates that identical instability equations give rise to Kluver's form constants whereby hallucinatory patterns arise via symmetry breaking (bifurcation theory). Hence, cosmic inflation, cancer stem cell differentiation, epigenetic transitions and hallucinatory patterns are analogous symmetry breaking systems.

They Ermentrout-Cowan model suggests that logarithmic retinocortical maps in the visual fields become oblique straight lines in the visual cortex V1. The stripes formed by hallucinogens (morphogens) are defined by Turing's animal coat pattern formation equations implying hallucinogen dynamics may be entirely defined through diffusion constants and chemical rate equations. The Turing structures can be replicated in a neural context with a *Mexican Hat distribution* representing the local excitations and lateral inhibitions. The threshold values of the field's peaks decide the inflection/critical point for morphogenetic (visual) field bifurcations. As long as the length scales or time frequencies of the activator-inhibitor gradients differ, stripes or spots can occur.

Geometric pattern formation during visual hallucination is a form of spontaneous symmetry breaking whereby an equilibrium system with certainly known initial conditions evolves to an unstable heterogeneous system with nonlinear wave equations (Bressloff et al., 2002). Moreover, Bressloff et al. confirmed these neuropsychedelics interactions in the cortical map can be simplified

EPIMETABOLIC REPROGRAMMING LANDSCAPE

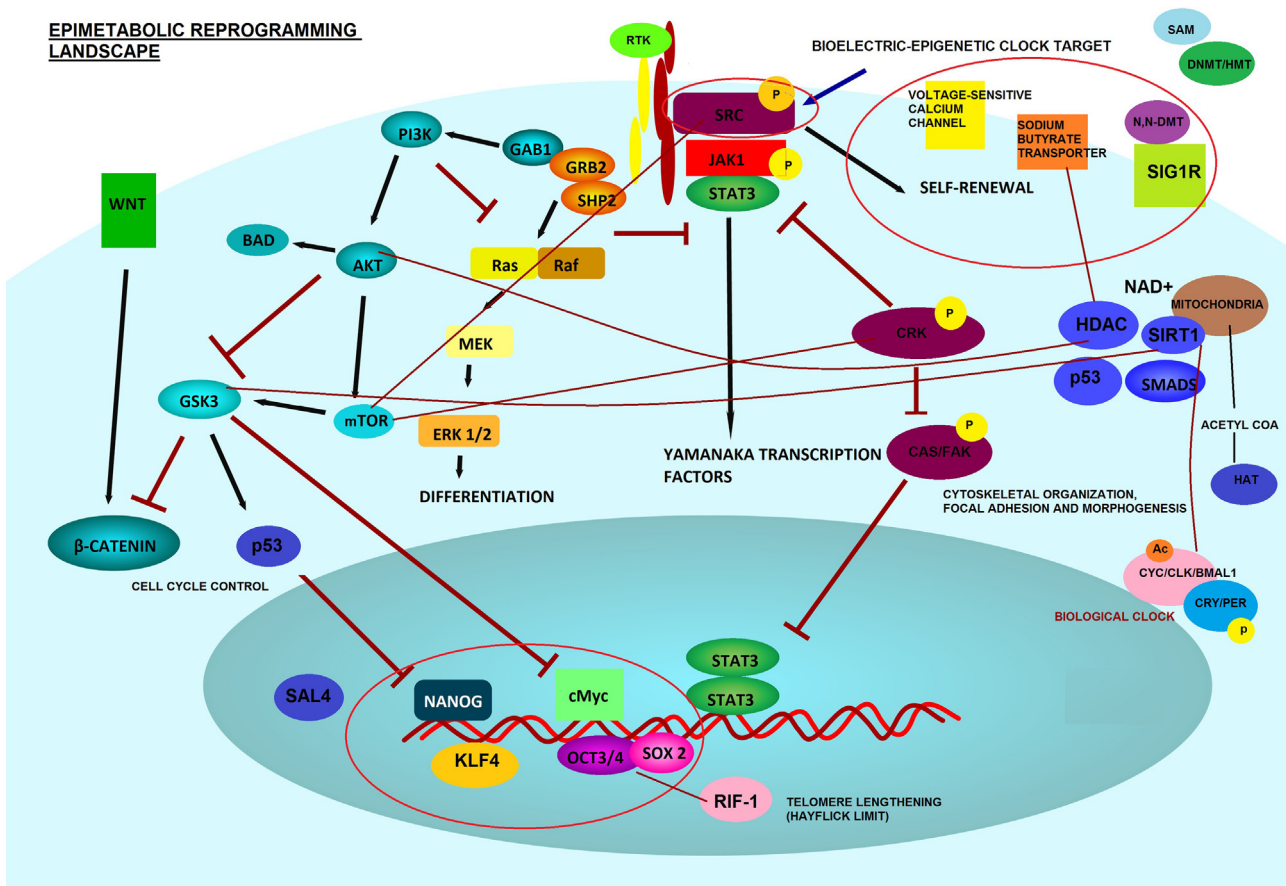


Fig. 8. Cancer Reprogramming Landscape: The discussed Src-mTOR-iPSC network and signaling cascades mapped as potential epigenetic cancer reprogramming pathways. Under the adopted quantum epigenetic landscape, cancer cell states are reversible (reprogrammable) phenotypes.

as a Mexican-Hat distribution, the topology of the epigenetic flux landscapes proposed by [Wu et al. \(2016\)](#) and [Wang et al. \(2008\)](#) discussed earlier. The Fourier-transform bifurcation plots of an eigenvalue $\lambda = -g + v\sigma\tilde{w}(q)$ where $\tilde{w}(q)$ is the Fourier transform of the Mexican Hat distribution representing the neural network lattice (energy landscape), q is the wavenumber, g is the decay or nonlinear coupling constant of neural interactions (vortex coupling constant), σ is the uncertainty measure (quantum potential Q) given an inhomogeneous field and v is the growth parameter of the spatial patterns.

Likewise, the [Coleman-Weinberg model \(1973\)](#) indicates that the symmetry breaking of a 4D relativistic scalar field fluctuation in the vacuum superfluid yields QED (Quantum electrodynamics) and the Standard model. Thus, gauge interactions and quantum particle fields are by-products of the vacuum energy fluctuations' self-organized symmetry breaking. The C-W model further states that a *Mexican-Hat landscape* is necessary for the spontaneous gauge symmetry breaking. In fact, the simplest relativistic model of scalar field spontaneous symmetry breaking is the Mexican Hat topology ([Goldstone, 1961](#)). Given the Lagrangian: $L = \partial^\mu \phi \partial_\mu \phi - V(\phi)$, the classical energy density of a discretized real scalar field $\phi(x)$ undergoing symmetry breaking is modeled as a harmonic oscillator: $E = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + V(\phi)$. These potential equations are identical to the Waddington flux landscape model discussed.

The pre-patterns of biological systems are Bohmian (pre-determined) developmental (morphogenetic) fields that have a non-uniform spatial arrangement of substances in a tissue (pre-programmed global structure where local peaks cause pattern formation) evolving through time by the influence of environmental fluxes. Then, the cell-matrix interactome's (epigenetic) energy

fluxes enable selective quantum adaptive evolution. Therefore, I propose that further studies into the effects of such endogenous psychedelics in cancer epigenetics and metabolism will unify the physical principles of space-time to energy vibrations in the biological domain. Moreover, above discussed signaling cascades provides the infrastructure for a cancer reprogramming epigenetic landscape (Fig. 8).

6. Conclusion

In conclusion, the cancer model developed in this paper asserts that the dynamical co-evolution of tumor cells and their microenvironment can be fully mapped by the energy fluctuations of its transcriptome-epigenome landscape. Cancer cells are not confined to the systemic orders and spatio-temporal boundaries non-cancerous cells are bound to. In theory, by manipulating the cancer development landscape through selective signal transduction based reprogramming, the malignant phenotype of the cells can be reversed towards an embryonic or default state regardless of the patient-specific driver mutations. The quantum formalism and the mathematical infrastructure of the reprogramming energy landscape have been developed hereby, where the epigenome is the central observer of the system and cancer cells co-evolve with their environment through quantum adaptive evolution.

Furthermore, I propose the bioelectromagnetic field of cells is coupled to its epigenetic flux rate. Hence, a bioelectric–epigenetic code is predicted to exist as the physiological counterpart of the pre-space quantum pilot wave guiding the turbulent energy dynamics of cancer cells. Moreover, key novel insights in epigenetic reprogramming pathways such as the Src-mTOR-iPSC networks,

endopsychedelics, energy metabolism pathways and bioelectric cascades have been discussed in support of the model.

Although mutagenesis is unpredictable, biological evolution is a quantum chaotic (deterministic) process whereby cancer cells have selectively adapted to their environments through the quantum evolution of signaling networks. The field of quantum biology has only few widely accepted empirical evidences and thereby the developed cancer model will extend its domain of research. Numerous experimental techniques to validate the presented quantum biological models have also been suggested. The pilot wave theory of quantum mechanics and the Hameroff-Penrose theory of decoherence are examples of physical models not within the mainstream sciences. However, the non-probabilistic framework of cancer evolution presented here displays an equivalent mathematical framework as the widely accepted *interpretations*. Moreover, I propose that measuring cytoskeletal dynamics is a direct measurement of epigenetic memory transmission. Environmental flux-specific chromatin remodeling (gene expression dynamics) and thereby epimetabolic regulation of cells determine cell phenotypes in response to environmental fluctuations, from which Natural Selection adaptively selects the preferred epigenetic (evolutionary) cell state. By treating the cancer cells and their environment as a single flux system, transgenerational epigenetic inheritance is the adaptive selection of preferred epigenetic (energy) landscape fluctuations in successive generations. I propose that such selective epigenetic inheritances are directly measurable by the cytoskeletal-matrix remodeling dynamics in cells for which cancer cells exhibit a more fluid, highly rapid and epimetabolically adaptive cytoskeletal-matrix rearrangements. In addition, I propose that by measuring the quantum dynamical properties and the vibrational frequencies of the hyper-chaotic flux BEC/Fröhlich condensation produced by cytoskeletal networks (e.g. measuring the phase evolution, vibrations and morphological properties of microtubules and actin through techniques such as interferometry, tip-enhanced Raman spectroscopy and AFM post-dielectrophoresis in relation to epigenetic fluxes of cytoskeletal-matrix remodeling signaling pathways in cancer cells such as the Src-mTOR-SIRT-c-Myc pathway), the QEH (pre-space Quantum Evolution Hypothesis) can be investigated and a quantum epimetabolome (interactome) can be conceived in biosystems.

Furthermore, I would like to propose fundamental changes required in clinical oncology to our physicians and medical policy makers. The following is a description of an ideal medical system with personalized (patient-specific) target therapies based on pharmacogenomics until a Universal cure is developed through the phenotypic energy landscape based reprogramming defined in this paper.

To begin with, a complete genome sequencing and epigenome sequencing of the tumor and healthy tissues must be assessed for all cancer patients to identify the mutations and its corresponding potential clinical targets (ex: over-expression/down-regulation of receptors, oncogenes, etc.). Our current system is limited to a screening window of selected few genes per patient and diagnosis is based primarily on a pathology report consisting of cytogenetic techniques such as Fluorescence in situ hybridization (FISH) karyotyping and basic immunocytochemistry as opposed to exome sequencing and epigenetic profiling.

In addition to the WGS (Whole Genome Sequencing), upon assessment of applicable tumor-specific assays, histopathology reports, biomarker expression profiles, etc., all identifiable molecular alterations (therapeutic targets) and signaling interactions should be mapped for the patient's cancer. Post- systemic analyses, all evidence based pharmacological agents, chemotherapeutic drugs and applicable interventions (ex: stem cell therapy, photodynamic therapy/optogenetics, immunotherapies, etc.) that could target the observed signaling anomalies (ex: upregulate/inhibit a

gene or its metabolite) should be assessed on the patient's tumor biopsy culture prior to clinical administration. Of course, this is considering that a tumor biopsy was attainable in the patient's cancer (majority of clinical cases).

Then, a cytotoxicity assay report graphically demonstrating the efficacy of each drug combination on the patient's biopsy culture must be used to determine the best treatment option. The target drug might not be within the class of drugs commonly used to treat the diagnosed type of cancer in clinical practice. For example, consider the case of Trish Keating (2015), a Vancouver colorectal cancer patient publicized in the media. The BC Cancer Agency's Personalized Onco-Genomics Program sequenced the patient's healthy and tumor DNA. A *blood pressure medication* was identified as an inhibitor of one of the identified tumor operating pathways. Post-two months of the identified target-specific clinical therapy, the patient went under complete remission while she couldn't with five years of invasive surgeries, chemotherapies and radiation therapy. This demonstrates the potential of personalized genomic sequencing based- clinical medicine.

To illustrate a clinical example of the above suggested system, consider a child with H3K27M mutant positive Glioblastoma Multiforme. Whole Genome-Epigenome Sequencing, tumor specific assays and immunohistopathology of the tumor and healthy brain biopsies must be performed. Using such personalized genomics data, the operating signaling pathways can be identified. Then, the efficacy of potential clinical targets (*regiment* chemotherapeutics and experimental drugs available) such as: panobinostat+GSKJ4, linsitinib, temozolomide, bevacizumab, tyrosine kinase inhibitors, carmustine, cisplatin, MGMT inhibitors, cell cycle checkpoint inhibitors, etc. should be assessed on the tumor cultures individually to compare their efficacies. The drug selection for the cytotoxicity-efficacy assay depends on the signaling pathway anomalies identified in the patient specific tumor (ex: EGFR, PTEN, mTOR, p53, etc.) rather than adhering to the pre-imposed conventional regime of surgery and/or radiation therapy.

For example, consider a child with grade I pilocytic astrocytoma undergoing chemotherapy cycles of vincristine. Post-few cycles, medical imaging showed no signs of tumor shrinkage. Hence, the regiment was changed to cycles of vincristine and carboplatin. It should be noted that such trial and error protocol repetitions are damaging to the child's immune system, healthy tissues (non-targeted cells/organs) and further increases drug resistance/aggressiveness in tumor cells. Meanwhile, preliminary (limited) genetic sequencing identified mutations in FGFR I, PTPN 11 and NF I. There are many targeted therapies for the MAPK/RAS/ERK pathway, MEK inhibitors, FGFR I inhibitors, etc. available clinically and experimentally for the identified mutations. For instance, IL-B08 and SHP099 are SHP2 phosphatase inhibitors that could target the recognized mutations. The personalized therapy candidates must be assessed on the patient's biopsy cultures along with the systemic regimen drugs (vincristine and carboplatin in this case) to compare the efficacy and determine the best drug(s) combination. At least, the anti-tumor efficiency of the identified personalized candidates must be compared with the efficiency of clinical regimens *in vitro*. If there are economic constraints, the option of private clinical care and costs to purchase the drug must be made available to patients.

Precision medicine based targeting patient-specific genomic-epigenomic mutations must be the global medical regime, where tumor DNA sequencing and cytotoxicity assays of identified drugs on tumor biopsy cultures must be assessed prior to clinical intervention. Targeted therapies may present unfavorable systemic side-effects. However, the same holds true for the non- specific clinically available regimens. I am only proposing the use of simple tissue culture methods on glass plates. Tumor graft models and *in vivo* animal studies are not even suggested. The outlined clin-

ical model also considers that breakthrough experimental drugs and non-conventional approaches are not made available when applicable nor informed to the patients.

To the biochemists and biophysicists, this model will be a guide to further advance experimental research in the field of Quantum biology and provide the mathematical foundations for cancer reprogramming with experimental implications. Lastly, to the mathematicians, the Universe is yours.

“If you want to find the secrets of the Universe, think in terms of energy, frequency and vibration.”– Nikola Tesla.

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